Request for Proposal

Timing the Throw in a Quick Knife Draw Competition

Praxis II

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1 Introduction

The following Request for Proposal (RFP) addresses Stryke Target Range, an indoor axe throwing, knife throwing, and archery club. The opportunity is primarily focused on a subset of knife throwing called quick draw, where the objective is to draw one's knife and hit the target faster than the opponent. The owners of Stryke are looking for a device that is able to assist a referee in quick draw competitions by indicating when to throw and measuring the time for each throw. They are also interested in implementing the device at Stryke to increase the enjoyment of the sport for customers and integrate it with their axe throwing and archery ranges.

Currently, there is no product specially designed device for knife throwing on the market. As of now, the IKTHOF uses a homemade timer made in Germany [B, 7] that has multiple critical flaws (specified in Section 2.3). The requested solutions will be referred to as an **LTD (light and timing device)** in this proposal. The intent of this RFP is to provide student design teams with the information required to prototype and produce a solution that addresses the quick draw timer requirements.

2 Background of the Sport and Setup

2.1 The rules of quick draw

Knife throwing is a competitive sport requiring the thrower to aim at a target from certain distances. In quick draw, speed and accuracy are tested as two throwers must hit a certain area of the target faster than the other.

The throwers, who are in separate lanes, stand in front of a set of 3 targets, though only the center target is used. A lighting system remotely controlled by the referee is placed between both sets of targets. After the referee initiates the timer, the light flashes green randomly between an interval of 2 to 5 seconds and the thrower must draw their knife from a sheath on their belt, see **Figure 2.1,** and hit the target in a specific ring as fast as possible. The targets used for knife throwing competitions consist of 5 concentric octagonal rings. A more detailed representation of the target is shown in **Figure 2.2**.

If both throwers' knives stick in the valid ring, the thrower with the faster time will win the round. If neither knives stick in the valid ring, it is considered a draw. For a set official rules by the International Knife Throwing Hall of Fame (IKTHOF), refer to Appendix A (Source Extracts), 1.

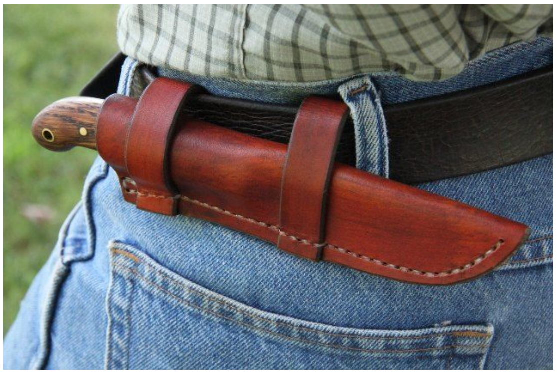


Figure 2.1 [A, 2]: A knife sheath is a cover for a knife. In competition, the sheath must be attached to one's belt [A, 1].

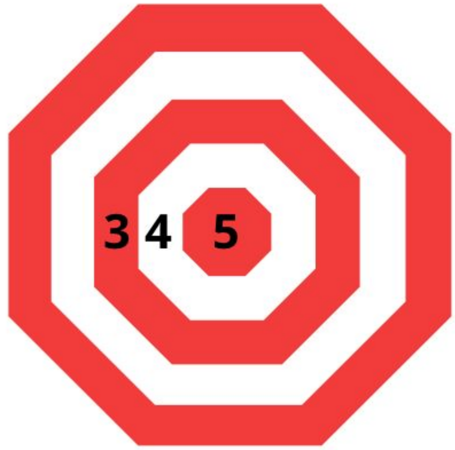


Figure 2.2 [A, 3]: In IKTHOF knife competitions, knives must hit rings 3, 4, or 5. Outside areas do not count. Before each round, throwers are told which ring to aim at.

The concern is that in competitions with high skilled throwers, the knives would hit the target within a fraction of a second from each other, making it impossible to tell by ear or sight who the winner is [Appendix B (Field Notes), 2]. A quick draw was demonstrated by two Stryke employees; the motion happened so fast that none of the members of our team could determine which knife hit the target first [B, 2].

2.2 Equipment

2.2.1 Outdoor set-up for competitions

Knife throwing competitions, which are held outdoors, use an LTD connected to an accelerometer attached behind the wall that the targets hang on. The force of the knife is enough to transmit through the wall to be felt by the accelerometer [B, 2].

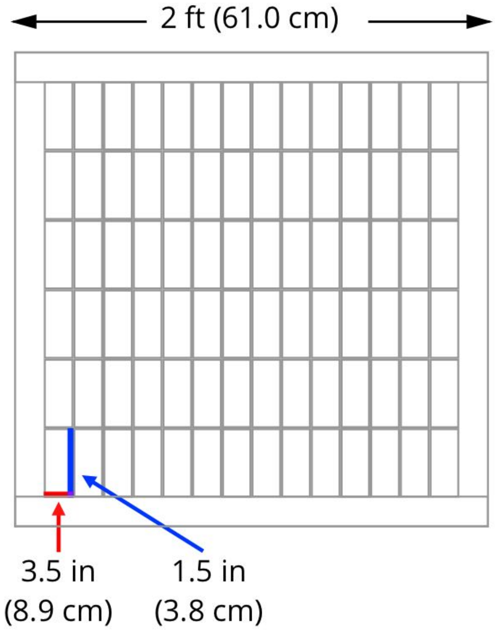
Due to competitions taking place in various outdoors locations, a portable device is needed so it can be assembled and disassembled on targets.

Figure 2.3 [A, 5]: The established system for quick draw shown at Finger lakes Knife throw competition.

The targets used for outdoor competitions, and in **Figure 2.3**, are comprised of 3 individual targets, however, only the one in the middle is used. These targets are called "end-grain targets", see **Figure 2.4**, as they are comprised of small cubes cut along the grain of a long wooden beam and then glued together so that the end grains are on the surface of the target. This allows for knives to stick in the wood easier than if the knife struck with the grain of the beam [B, 2]. For the remainder of this proposal, end-grain targets meant for outdoor competitions will be referred to as **outdoor targets.**

Figure 2.4 [A, 6]: Zoom in of the target. Note the grains make up the surface of the target.

Figure 2.5 is the target used in competition that Stryke made for our team and **Figure 2.6** is its measurements.



Figures 2.5 (left) [A, 6] and 2.6 (right) [A, 3]: Image and measurements of an outdoor target. The target is 4.15 in (10.5 cm) thick. The diagram is to scale.

2.2.2 Indoor set-up for throwers

Unlike outdoor competitions, indoor venues consist of individually caged lanes, as seen in **Figure 2.7**. There are lines on the ground to indicate 3, 4, and 5 meters away from the target, see **Figure 2.8**.

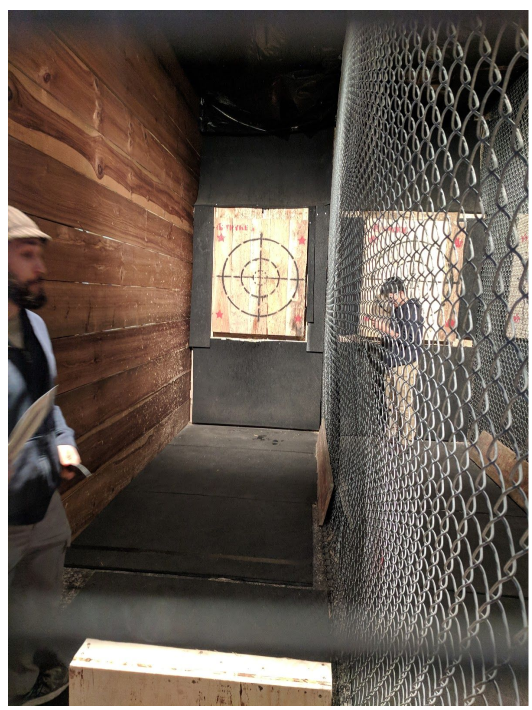


Figure 2.7 [A, 6]: View down a lane at Stryke.

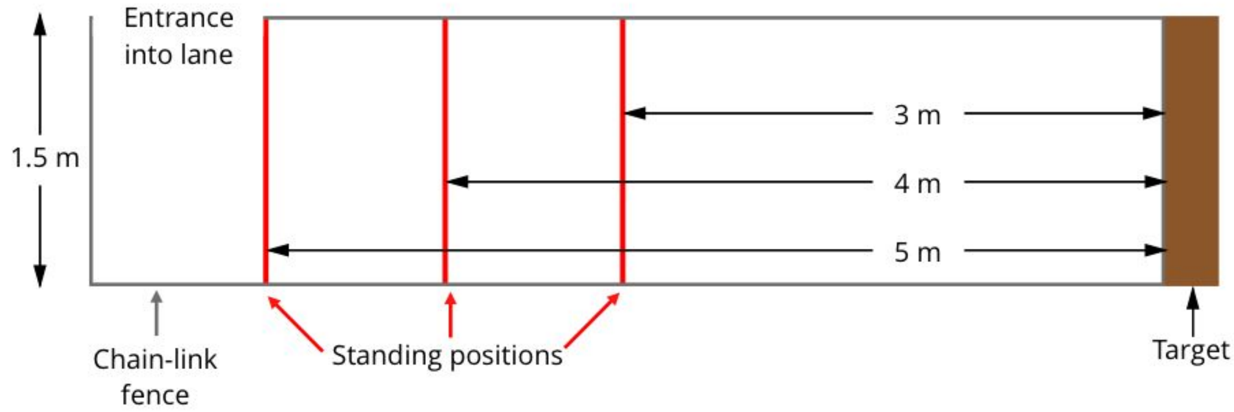


Figure 2.8 [A, 3]: Scaled layout of each lane at Stryke. Players can throw from any distance beyond 3 meters from the target but 3, 4, and 5 meters are marked on the ground for reference.

As well, the construction of the targets meant to be used indoors is different than construction of outdoor targets. The indoor target is comprised of 5 wooden planks placed vertically next to each other and then screwed to a frame; the planks make up the surface of the target and is spray painted with a bulls-eye, see **Figure 2.9**. When the front planks need to be changed due to damage from knife hits, all 5 planks are unscrewed and replaced; the frame gets reused. For the remainder of this proposal, these targets used at Stryke will be referred to as **indoor targets.**

Constructing the indoor targets requires less manual labour than the outdoor targets, as the planks can be easily replaced while the outdoor targets require the builder to cut up the wood for the fibres of the wood to be perpendicular to the surface of the target. Indoor targets are hooked to the wall with french cleats [A, 7 & 8], meaning that they can be manually removed by just lifting the plank up so it unhooks from the wall.

Figure 2.9 [A, 6]: Image of an indoor target at Stryke. The board is 4 ft (121.9 cm) tall and 3 ft 3 in (99.1 cm) wide.

Figure 2.10 [A, 6]: Image of a french cleat connected to the frame of the indoor targets.

2.2.3 The knife

According to the official rules by the IKTHOF, the knife used in competitions must be a knife that is at least 12 inches long and at least 12 ounces in weight. The knife must be drawn from a sheath on the thrower's waist [A, 1].

The common parts of a knife are listed below with respect to **Figure 2.11** [A,9]**:**

1. The blade
2. The handle (where you hold the knife typically)
3. The point (tip of the knife used for piercing)
4. The edge (cutting surface of the knife extending from the point to the ricasso)
5. The grind (cross-sectional shape of the blade)
6. The spine (thickest section of the blade--typically in the middle for two-edged knives, and towards the top for single-edged knives)
7. The fuller (the groove added to make the blade lighter)
8. The ricasso (The flat section of the blade located right before the handle)
9. The guard (barrier between the blade and the handle to prevent the hand from slipping forwards onto the blade)
10. The hilt or butt (end of the handle)
11. The lanyard (not typical of throwing knives)

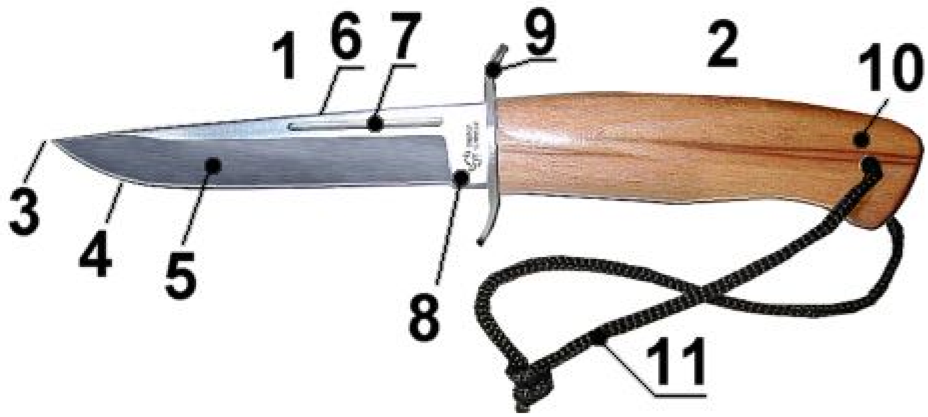


Figure 2.11: Diagram showing parts of a typical knife [A, 9].

The knives typically used in competition, shown below in **Figure 2.12,** are simpler in terms of number of different components, having only a blade, edge, handle, spine, fuller, point and hilt. The knives, except the black one in **Figure 2.12** also show another feature of hollow circles that can assist in gripping the knife.

Figure 2.12 [A, 6]: 4 sets of knives used at Stryke. From left to right: 8 inch knife, 9 inch knife, 13 inch knife, 12.5 inch knife. The 8 and 9 inch knives are meant to be used by beginners and the 12.5 and 13 inch knives are meant to be used in competitions.

2.3 Issues with the current timing mechanism

Based on the lived experience of the employees at Stryke, our team has learned that there are a number of issues with the current system that affect the quick draw community.

2.3.1 Causes throwers to change line of sight

The current lighting system used in IKTHOF quick draw competitions is placed high up between the targets and is not in the throwers' direct line of sight. Some lanes are far enough apart that the light is not in the peripheral vision of the thrower, requiring both throwers to rotate their heads to see the light, hindering the speed of their reflexes [B, 2]. A throwers can put their knife sheath wherever they prefer on their belt. Because of this freedom, under certain initial starting positions, the thrower may be put at an unfair disadvantage. For example, if a thrower must look to their right to see the light, is right handed, and prefer to have their sheath on their left side, their sheath may not be within their peripheral vision, putting the thrower at a disadvantage compared to other knife throwers.

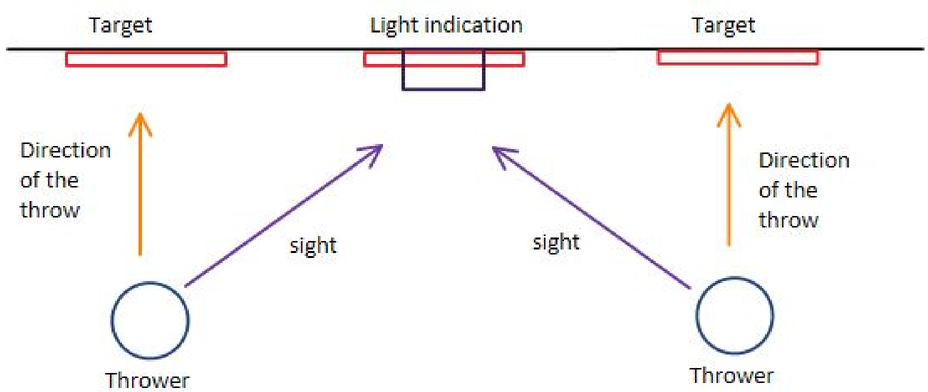


Figure 2.13 [A, 3]: An illustration of the current system. Note the inconsistency between the line of sight and the direction of the throw.

2.3.2 Screen overheating

An issue with the timer display is that the LCD screen gets damaged when exposed to the sun, causing the crystals to potentially overheat or burn, leaving the screen blank [B, 2]. When LCD screens are exposed to high UV radiation it "damage[s] the long chain organic molecules that make up liquid crystals, causing the chains of molecules to break down" which then would cause the screen to not display any image [A, 10]. This is a result from the competition taking place in the outside environment where there is heavy exposure to UV radiation [A, 11].

2.3.3 Precision

During competition, if the two targets share the same wall, the vibration felt by one knife hit may be transmitted through the entire wall. Both accelerometers may interpret the force, posing the problem of incorrect timing measured for the other device [B, 2].

3 Engineering Opportunity

The engineering opportunity proposed by Stryke consists of a LTD designed specifically for IKTHOF quick draw competitions. Stryke wants the LTD to have a light system above both targets in each lane to be able to light up at a random time between a 2 to 5 second interval, and to be able to indicate which knife successfully hits the target first. In addition, Stryke wants the LTD to be able to display the time taken from the moment the device lights up the "Start" light, to the moment the thrower's knife reaches the target clearly, indoors and outdoor. Indoors, the thrower may throw from any point between 3 to 5 meters, while for outdoor quick draw competitions, the minimum distance a thrower may stand is 3 meters [B, 2 & 4].

4 Stakeholders

The following stakeholders were sorted into two categories, primary and secondary, after an assessment using the definitions of power, impact, interest, and influence from *Managing Project Stakeholders* [A, 12]. Refer to [Appendix C (Engineering tools), 1] for further clarification of the roles of each stakeholder.

Primary Stakeholders

4.1 Stryke Target Range

Stryke needs a mechanism to use in their future competitions that will allow throwers to be at equal starting conditions. Not only can the mechanism be used to simulate competition conditions, but it can also create a more competitive and entertaining environment for Stryke customers to enjoy [B, 2].

4.1.1 General Employees

The general employees at Stryke have many responsibilities which include cleaning, making boards, coaching, and troubleshooting the devices and equipment used at Stryke [B, 2]. With the creation of a LTD that satisfies their needs, the general employees at Stryke would have another task to learn how to maintain and operate.

4.1.2 Manager

There is one manager at the Dundas Stryke location, Ryan [B, 1]. As the manager, Ryan has administrative tasks to adhere to [B, 2]. In the chance that Stryke would like to make profit from the opportunity, Ryan would be burdened with the task of advertising and selling the product (or create a new position at Stryke to handle profit related to advertisement of the LTD, and perhaps other future products).

4.2 IKTHOF

Opened in September 2003, the IKTHOF (International Knife Throwing Hall of Fame) is a hall of fame for knife and axe throwing whose mission is “To Honor, Preserve, Educate, and Promote” [A, 15]. The IKTHOF holds yearly quick draw competitions for the knife throwing community (and axe throwing community). Currently their timing mechanism is flawed as specified in Section 2.3. If a team is able to produce a solution that performs better than their current design for quick draw timing, then the IKTHOF may choose to adopt the proposed solution(s), therefore requiring the IKTHOF to consider some changes to their regulations and equipment.

Secondary Stakeholders

4.3 Competitive and professional knife throwers

There exists many categories of competitions for knife throwing that range by distance, categorize by skill, and test for speed and accuracy [A, 15]. To practice for quick draw, it would be ideal to simulate a real competition and a good timer meant for knife throwing. This would ideally improve the results achieved by throwers in quick draw competitions.

4.4 Local throwing community

As the popularity of knife throwing for kids and their parents grows, Stryke is currently trying to set up a children's outreach program that will provide kids with proper training as an after-school event [B, 2]. Other axe throwing clubs such as Bad Axe Throwing offer special discounts for events such as birthdays [A, 13].

Establishing a timed system will provide an opportunity for the community to go head-to-head in a physical skill based activity under a safe but competitive environment [B, 2] and promotes concentration, focus, and increases reflexes [A, 14], [B, 2].

This system can be used by regular customers, since they are part of the local throwing community and they would also benefit from a more dynamic and competitive environment.

4.5 Competition Audience

Knife throwing is a spectator sport; with the competition audience looking at the LTD, their enjoyment (or dissatisfaction) of the quick draw competition can be influenced by the presentation of the LTD.

5. Engineering Requirements

A further justification of the requirements is provided in Section 6.

Define

- "Start" is when the player is allowed to throw their knife, which will be indicated by the lighting system.
- "Stand-by" is when the player should be ready for the "Start".
- "Visibly Clear" with respect to a normal person — a person with 20/20 vision [A, 16].

5.1 High Level Objectives (HL)

1. Develop a viable method of tracking the time taken from when the "Start" light has been indicated to the moment when the knife reaches the target, for both indoor and outdoor ranges (as requested by primary stakeholder, Stryke [B, 2]).
2. Develop a viable method to clearly indicate to the throwers when to be on "Stand-by" and when to "Start" [B, 2].
3. Address the problems of the current LTD; refer to issues with the current system in Section 2.3.

5.2 Detailed Objectives (DO)

1. Track the time between “Start” and the knife hitting the target with accuracy.
2. Find a solution that is portable.
3. Find a solution with a lighting system that is visibly clear for throwers.
4. Find a solution that is able to function reliably under both indoor and outdoor conditions.
5. Produce a solution that is power conservative.
6. Affordably implement the device at Stryke and IKTHOF quick draw competitions.

5.3 Metrics

1. The absolute value of uncertainty of the timer (in seconds) (HL1).
2. Portability (HL2)
    a. Weight of all parts of the device in kilograms (kg).
    b. Dimensions in cubic centimeters (cm × cm × cm).
3. Visibility (DO1)
    a. Lux level of the light.
    b. Size of display in terms of character height (mm).
4. Durability (DO2)
    a. Level of risk of the LTD being disabled or damaged by a knife strike. Components of LTD being considered are the lights, wires and time display, and all other key components.
        i. **Ideal:** The LTD will not experience the impact of a knife being thrown during operation, or can withstand the impact of a knife being thrown during operation.
        ii. **Good:** Low risk of LTD being disabled due to the impact of a knife being thrown during operation.
        iii. **Unacceptable:** High risk of LTD being disabled due to the impact of a knife being thrown during operation.
    b. Time the system is able to function under continuous usage
        i. **Ideal:** Can function for more than 3 consecutive hours.
        ii. **Unacceptable:** Cannot function for 3 consecutive hours.
5. Power consumption of the device in watts (DO3).
6. Cost of materials in Canadian Dollars (DO4).

5.4 Constraints

1. Must be able to indicate the time difference from the "Start" light turning on, to the knife hitting the target.
2. There must be a LTD for each individual target. Highly preferred to be placed above the target.
3. The LTD must be able to compare the time for 2 different lanes of knife throwing [B, 2].
4. Must have the difference of reaction time between the two timers be within the uncertainty of the timer.
5. Must be a time display above both targets to show what time each thrower took for their knives to hit the target.
6. The timer must record the time to at least the nearest hundredth of a second [B, 2].
7. Must be able to assemble and disassemble over indoor and outdoor targets.
8. Must be compatible with the current setup of Stryke.
9. The lighting system:
    i. Must be directed along the same path as the direction of the throw.
    ii. Must not distract the throwers.
    iii. Must use a green light to indicate "Start".
    iv. Must indicate "Stand-by". Yellow or amber coloured light is highly prefered.
    v. Must be able to indicate "Start" within a 2 to 5 second randomly after the "Standby" light has lit up.
    vi. May be able to indicate which knife hit the target first.
10. The timer must start recording when the light indicate "Start", and stop when the knife hit the target.
11. Must be sun resistant.
12. May be water repellent.
13. Must not have a rating of **unacceptable** for the metric of "level of protection for wires and light".
14. Must be able to use the LTD both indoors and outdoors.
15. Must be compatible with a portable power supply.
16. Should not rely on the usage of a power outlet.
17. Must provide at least one indicator light (indicating "stand-by", "start") that can be easily recognized by a person with normal vision (20/20 vision, [A, 16]) at 5m away from the target under daylight conditions (with daylight having a lux level of approximately 10,752 lux) [A, 19].

18. The individual numbers of the time displayed must have a character height greater than 38mm [A, 18] The width of the total time display must be smaller than 1.5 m, which is the width restriction of a lane in Stryke, referring to Section 2.2.2.
19. The material and construction cost of 2 LTDs must not exceed $900 [B ,2].

5.5 Criteria

1. Prefer smaller uncertainty of the timing of the device.
2. Prefer lighter weight.
3. Prefer smaller dimensions.
4. Inverted U-shape for lux level.
5. Complex-utility shape for the character height of the time display (refer to 6.5.5 for details of the utility shape)
6. Prefer **ideal** in terms of the level of risk of the LTD being disabled or damaged by knife strike over **good**, and **good** over **bad.**
7. Prefer **ideal** level of ability for LTD to function under continuous usage.
8. Prefer lower power consumption.
9. Prefer lower materials and construction cost.

6 Justification of Requirements

6.1 High Level Objectives

The high level objectives are justified by our interactions with the primary stakeholder, Stryke, and issues with the current system in Section 2.3. For more details, refer to Section 5.1.

6.2 Detailed Objectives

DO 1: Track the time from the start of the knife being thrown, and the knife hitting the target with accuracy

This objective is essential since time is a deciding factor in competition in quick draw [B, 2], and a system with higher accuracy can legitimize the method to be used in competition, contributing to High Level Objective 5.1.1 of developing a method of tracking the time of throw.

DO 2: Find a solution that is portable

The LTD needs to be capable of functioning in both indoor and outdoor ranges, as it will be used in both outdoor competitions and indoor ranges. Therefore it must be relatively easy to transport, as requested by a primary stakeholder, Stryke [B, 2].

DO 3: Find a solution with a lighting system that is visibly clear for throwers

The lighting system would need to have a lux level which will not interfere with the throwers but will not be so low that they would have trouble seeing the light during the game/competition.

DO 4: Find a solution that is able to function reliably under both indoor and outdoor conditions

Meaning, the device should be able to handle radiation from the sun amongst other variations of radiation without becoming overheated or over energized, as this has been a problem presented by the primary stakeholder, Stryke, and is needed to be addressed as per high-level objective 5.1.3 [B, 5].

The key components of the LTD may be exposed to knife strikes. Minimizing the risk of disabling the LTD under the impact of a knife will contribute to the durability of the LTD, allowing for reliable functioning. This falls directly under high level objectives 5.1.1 and 5.1.2 as those objectives cannot be achieved with a dysfunctional device. Similarly the LTD must also be able to function under continuous usage, adhering to 5.1.1 and 5.1.2.

DO 5: Produce a solution that is power conservative

Due to the request by Stryke to create a solution that is not dependent on a power outlet, the power supply must be portable such as a battery and other means [B, 4]. With the power supply being a portable power supply, to reliably record time and light up as programmed, the power supply must be able to continuously function. To decrease the chance of the LTD power supply failing during a competition, it is an effective way to address this problem by providing a power conservative solution that can significantly reduce the chance of the LTD power supply failing, adding to the reliability, contributing to high level objectives 5.1.1 and 5.1.2.

DO 5: Affordably implement the device at Stryke and IKTHOF quick draw competitions

Based on the interaction with primary stakeholder Stryke, there exists a limited amount of resources to be devoted into implementing an LTD at Stryke [B, 2]. Therefore for the proposed solution to be viable as suggested by high-level Objectives 5.1.1 and 5.1.2, the solution must be affordable to be implemented, while also complying with IKTHOF competition rules.

6.3 Metrics

Metric 1: The absolute value of uncertainty of the timer (in seconds)

To evaluate the accuracy of the LTD as specified in Objective 5.3.1, data regarding the time accuracy can be acquired through the specification of the timer used in constructing the LTD. If the device is custom made, then there must be a viable method of experimentally finding the uncertainty of the timer.

Metric 2: Portability

By following the definition of a portable device as mentioned in [A, 20], a portable device is a device that is small (dimensions in cm x cm x cm) and lightweight (weight in kg).

a. **Weight of all parts of the device in kilograms (kg)**
   The weight of the device can be measured using a scale.

b. **Dimensions (cm x cm x cm)**
   Measure the dimensions of the LTD using a bounded box. A bounded box is defined as the smallest rectangular prism that can fully circumscribe the dimension of the LTD, of which is stored in a manner that will not cause permanent deformation nor damages. It is used to evaluate the dimension metric since it would represent the possible space the LTD may occupy when stored.

Metric 3: Visibility

a. **Lux level of the light**

   The lux level is used in order to determine what the brightness of the new light system is. The lux level is a measure of illuminance in foot candles (lux), which is the light density per square foot [A, 19]. To find lux levels, the lux meter on mobile phones is the most accessible to people, as can be seen in [A, 26]. Given that 95% of the American population owns some sort of smart device [A, 27], it is the most accessible method of finding lux levels.

b. Size of display in terms of character height (mm)

The display indicates the time it takes from “start” to when the knife hit the target. The display is to be visible to all throwers and in order to test its visibility, the relationship in [A, 18] can be used to determine the minimal character height that will allow minimally sufficient character recognition. In practice, a more reasonable lower bound for the distance of the audience during a quick draw competition from the target is 15 meters, meaning that the clear readable character height must be at least **38 mm** [A, 18].

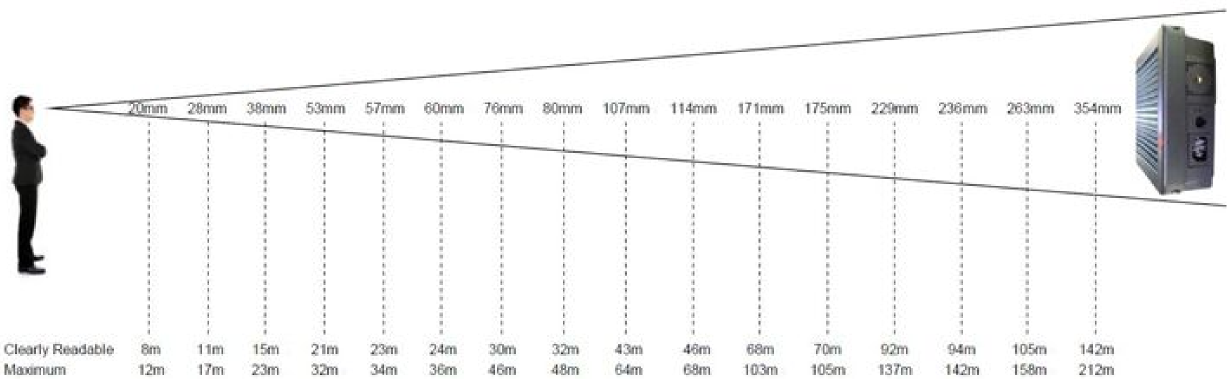


Figure 6.1 [A, 18]: Through experimentation these were the optimal character height at certain distances [A, 18].

Metric 4: Durability

a. Level of risk of the LTD being disabled or damaged by knife strike, which is consisted of the lights, wires and time display, and all other key components.

   i. **Ideal:** The LTD will not experience the impact of a knife being thrown during operation, or can withstand the impact of a knife being thrown during operation.
   ii. **Good:** Low risk of LTD being disabled due to the impact of a knife being thrown during operation.
   iii. **Unacceptable:** High risk of LTD being disabled due to the impact of a knife being thrown during operation.

The risk of LTD being damaged by the impact of the knife directly correspond to the durability/reliability of the system, relating to Objective 5.2.4.

b. Time the system is able to function under continuous usage
   i. **Ideal:** Can function for more than 3 consecutive hours.
   ii. **Unacceptable:** Cannot function for 3 consecutive hours.

The competition usually lasts about 2-3 hours [B, 4], and therefore the LTD must remain functional continuously during that time to be durable/reliable as mentioned in Objective 5.2.4.

Metric 5: Power consumption of the device in Watts

The power consumption is the amount of energy used by the device over a period of time. It is important to evaluate the level of power conservativeness of the device. It is important to use devices which utilise little energy so that they can be taken to various events and not depend on the source of energy other than a battery. Therefore this metric is eligible to be used to evaluate the power conservative of the LTD in 5.2.5.

Metric 6: Cost of materials in Canadian Dollars

To measure the material and construction cost of the LTD; it would be ideal to use Canadian Dollars as Canada is where the Stryke locations are located.

6.4 Constraints

Most constraints are justified by our primary stakeholder Stryke [B, 2, 3 & 4] and relevant IKTHOF rules [A, 1]. For other constraints, the justifications are listed below:

Constraint 4: Must have the difference of reaction time between the two (or more) timers be within the uncertainty of the timer

For Metric 5.3.1. involving the uncertainty of timer to be valid, the time difference.

Constraint 12: May be water repellent

As specified in [B, 6] the manager of Stryke said that it is not necessary but it would be useful to produce a water repellent solution.

Constraint 13: Must provide at least one indicator light (indicating "stand-by", "start") that can be easily recognized by a person with normal vision (20/20 vision, [A, 16]) at 5m away from the target under daylight conditions (with daylight having a lux level of approximately 10,752 lux) [A, 19]

The indicator light should be visible under both indoor and outdoor condition. The light should still be clearly visible in daylight at 5 meters away, since this is the worse case scenario due to a decreasing of lux with distance [A, 29] and bright sunlight making it harder to see [B, 2]. If this constraint is satisfied, then the display can be read clearly under most circumstances.

Constraint 18: The individual numbers of the time displayed must have a character height greater than 38mm [A, 19] The width of the total display must be smaller than 1.5 m, which is the width restriction of a lane in Stryke, referring to Section 2.2.2.

The character height must be larger than 38mm since that is the minimum character height that would be visible from 15m. Although in most competitions the distance from a thrower to his/her target is less or equal to 5m [B, 2]. 15m is chosen so that it guarantees that the competitors, the referee and the audiences can see the time display clearly.

6.5 Criteria

Criteria 1: Prefer smaller uncertainty of the timing of the device

The smaller the uncertainty of the LTD, the higher the accuracy of the time displayed, the better we adhere to track the time with accuracy in detailed objective 5.2.1.

Criteria 2: Prefer lighter weight

With a lighter weight, it would require less force and effort to carry the LTD from one location to another location, say from an indoor range to an outdoor competition. With less force and effort to carry an object, we can then conclude that if you keep every other factor constant and solely adjust the weight of a device such as a LTD, then the lighter a device is, the more portable the device is.

Criteria 3: Prefer smaller dimensions

Smaller dimensions would mean that the device would take up less room therefore making it easier to carry and store the device. In an article by Graham Pitcher, engineering designer, he states that the smaller the device the more portable it tends to be, therefore, it is preferred that there is a smaller device [A, 32].

Criteria 4: Inverted U-shape for lux level

Based from an experiment completed, the light levels which would impose the most concentration for when participating in the quickdraw match was around 550 lux levels [A, 19]. This would prove effective with the quick draw participants because a light which is flashing at around that level would rate at around a score of 1.25 [A, 21].

U-shape for lux level due to sudden increase in the lux level should have an optimal point where the lux levels would be best for the thrower. However, if this point is not reached then the lux levels may be too low.

Criteria 5: Complex-utility shape for the character height of the time display

The characters height of the text on the timer should be at least 38mm for each character, referring to Section 6.3.3b. The utility of the display increase as the character height increases, since it would be easier to identify over the same distance [A, 18]. As the height become larger, most people will be able to identify the character, meaning the utility will only increase slightly, up to a point the display is not compatible with the setup with Stryke, and utility drop to 0, as shown in **Figure 6.2**

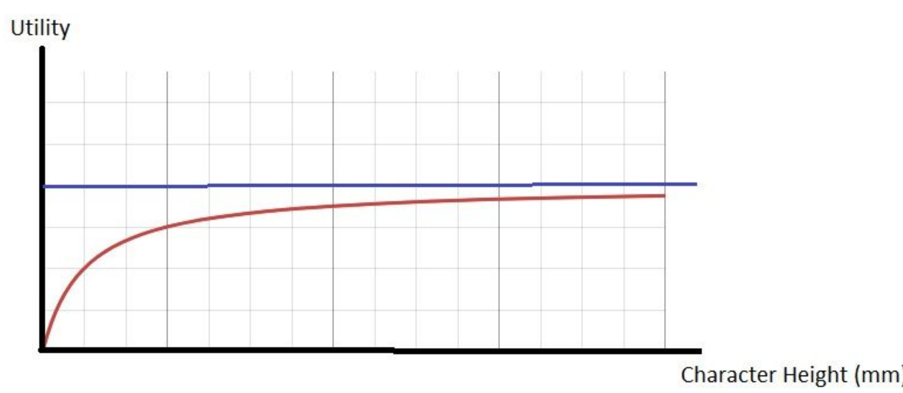


Figure 6.2 [A, 3] The utility-Character Height graph for time display. Blue line represents a horizontal asymptote, and the red line represents the utility-height relationship, and the black lines represent the axes.

Criteria 6: Prefer *ideal* in terms of the level of risk of the LTD being disabled or damaged by knife strike over *good*, and *good* over *bad*

The lower the risk of LTD being damaged or disabled, the more reliable it is at remain functioning, the more durable it is at resisting or preventing knife strikes from damaging the LTD.

Criteria 7: Prefer *ideal* level of ability for LTD to function under continuous usage

The longer the LTD is able to function under continuous usage, the more durable it is with at satisfying objective 5.2.4, making it a more preferable design.

Criteria 8: Prefer lower power consumption

A lower power consumption, assuming other factors are unaffected would allow for a lower material cost as you wouldn't have to replenish the portable power supply as frequently. Subsequently, the justification of preference of a lower material cost follows directly after.

Criteria 9: Prefer lower materials and construction cost

As mentioned by an employee of Stryke, who is our primary stakeholder, a lower materials and construction cost is preferred [B, 2].

7 Reference Designs

7.1 Cup stacking timer mat

Cup stacking, also known as speed stacking, is a sport where a player must position plastic cups in a specific arrangement in the shortest time possible. Cup stacking competitions use a special touch-sensored mat, see **Figure 7.1**.

7.1.1 How the mat works

The touch activated sensor, referred to as the *trigger*, is activated by pressure, such as a hand lightly pressing it, and begins and ends the timer [A, 23], see **Figure 7.2**. As in **Figure 7.3,** an LED screen readers out the time beside the mat, although some models have the timer and mat embedded together. The mat is made with a non-slip texture such as foam allowing for it to be light, and can easily be rolled up and stored away. This product is patented by Speed Stacks and is the official timing device of the World Sport Stacking Association (WSSA) [A, 23].

Figure 7.1 [A, 22]: Model with a mat, embedded timer, and trigger.

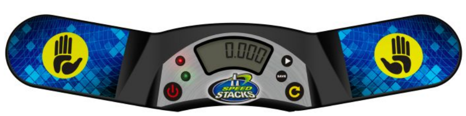


Figure 7.2 [A, 22]: Large LED read out meant for tournaments

Figure 7.3 [A, 22]: Model without a mat and only the timer and trigger.

7.1.2 Issues with the mat

The device deals with uncertainty [A, 23], is light and small, and is affordable (every item in SpeedStack's store is less than $100 USD [A, 22]). However, this device would not withstand impact of a knife. Elements of this device could be implemented in a proposed LTD.

7.2 Target Impact Signal System (TISS)

Furthermore, implementations of a LTD have previously been designed such as the Target Impact Signal System (TISS).

7.2.1 How TISS works

This device utilizes a metallic plate with a sensor that can handle projectiles, specifically bullets, that can travel speeds up to 1333 km/h [A, 24], seen in **Figure 7.4**. An timer add-on can allow for the TISS to measure how long it took for the projectile till it hit the target, shown in **Figure 7.5**. The light will flash first indicating start, and will flash again when the projectile hit the target, indicating hit.

7.2.2 Issues with TISS

To utilize the system for knives would be undermining the initial purpose of the device. Knives travel at speeds of up to 60 km/h (about 5% of the speed of a bullet) [A , 25], which means that this system is not fully compatible for knives. As well, the TISS uses a strobe light, which is a light source that is visible at 1000 meters with the naked eye [A, 28], while in quick draw, the player throws from 3 to 5 meters away from the board. One major drawback of this system is that the target is made of a steel plate [A, 28], making it incompatible with knife. In quick draw, the knife have to stick in the target in order to be valid [B, 2], and this would be impossible with a steel plate as target. Therefore this system is incompatible with quick draw.

Figure 7.4 [A, 30]: Metal plate that acts as the target.

Figure 7.5 [A,28]: This timer can be used for the measure of how long each

person takes to hit the device and it acts as an extension to the already existing target system created by TISS.

7.3 Fully Automatic Timing System with line-scan camera

Apart from the gun shooting community, sports communities such as track and field also have their own method of tracking the precise time when the runner crosses the finish line, and an example of this is the FinishLynx Fully Automatic Timing System (F.A.T.).

7.3.1 How the camera works

This system utilize a line scan camera to determine which athlete cross the finish line first with a high precession of 1/1000 second. It consists of three main components: a photo finish camera at the finish line, a laptop with the corresponding software, and a start sensor indicating the start of the match, showing in **Figure 7.6**. The camera will record the video of athletes crossing the finish line with 1000 frame per second accuracy, and the software can determine the chest position of each athlete in each frame, showing in **Figure 7.7.** The start sensor can be attached to the starting pistol indicating start for the timer, while athlete's chest line crossing the finish line will indicate stop, therefore tracking the performance time for each athlete [A, 31].

7.3.2 Issues with the camera

One of the major drawbacks of this system is that it is very expensive, ranging from $4995 to $24995 for different variations [A, 33]. This vastly exceed the cost constraint of less than $900 in Section 5.4, violating the cost constraints of the system, therefore making it impractical to be implemented to knife throwing communities.

Another implication of this design is that there is no protection for the line camera system to prevent it from being damaged by the knife, meaning that it rates very low in terms of durability, making it not an ideal system to be implemented from quick draw.

Figure 7.6 [A, 31]: Equipments used by the Lynx automatic timing system, including the line scan camera, corresponding software, and other components.

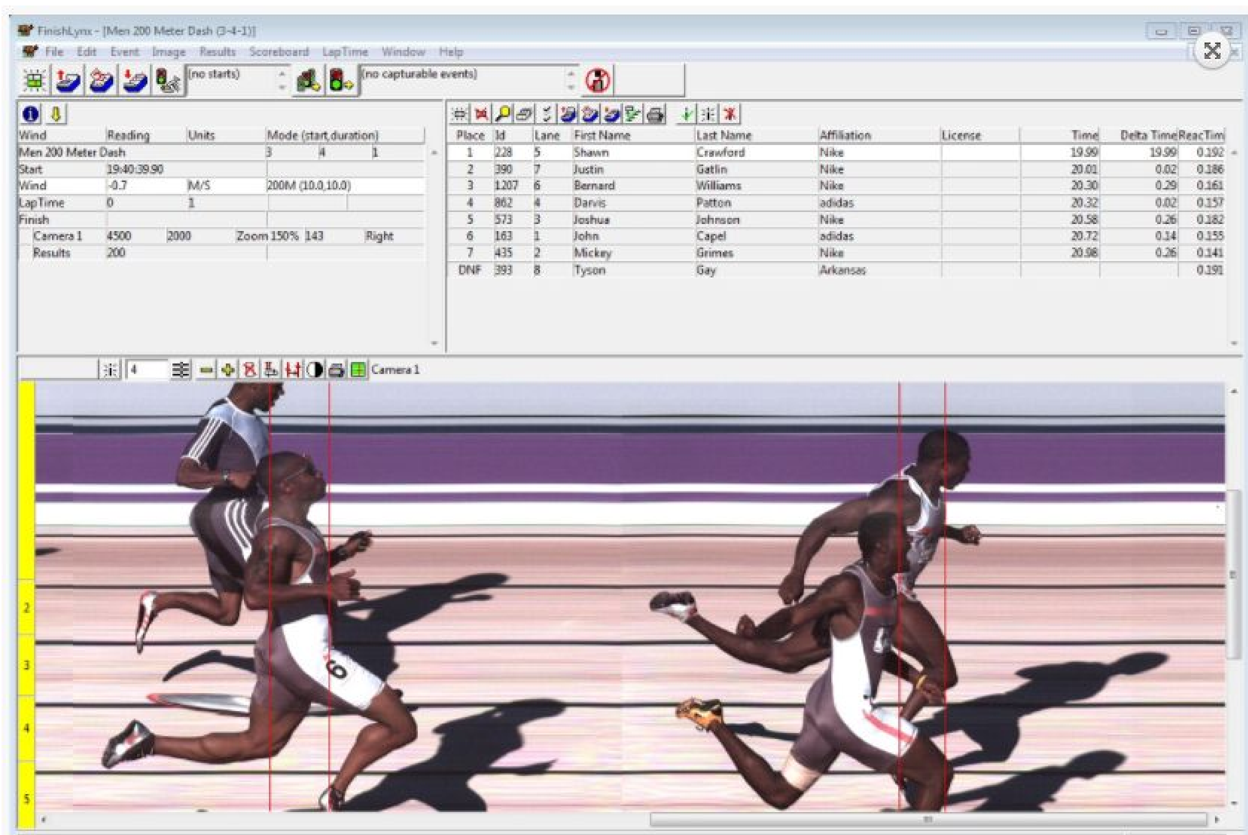


Figure 7.7 [A, 31]: The software identifies the chest line for each athlete, which will be used to measure to performance time for each athlete.

7.4 Aquatic Timing System

Timing is essential for many sports, and this also applies to competitive swimming. Similar to the timing system with line-scan camera in Section 7.3, the aquatic timing system will measure the time it takes for each athlete to swim a certain number of laps.

7.4.1 How the timing system works

The device mainly consists of four components: a sound system to indicate start; relay take-off platforms that can identify if there are any athletes started early, making sure that it is fair for all competitors; touchpads attached to the end of lane that can generate a signal when pressed, shown in **Figure 7.8**; and a timing console that processes the information [A, 36]. More detailed requirements of the system for international competition is provided by FINA [A, 37].

The major difference of the aquatic system is the use of touchpad instead of a camera to record the time. An example of this reference design is the Daktronics Touchpad T-7000s Series shown in **Figure 7.9.**

Figure 7.8 [A, 34]: Showing the touchpads secured on the end of the lane during competition.

Figure 7.9 [A, 35], A front image of a Daktronics T-7000 series touchpad

7.4.1 Issues with the timing system

One major drawback of the design is the dimensions and weight of the system. Each touchpad has a dimension ranging from 559 mm x 1524 mm up to 914 mm x 2413 mm, and weights from 12.8 kg to 32.9 kg [A, 35]. A system with multiple boards along with the control console would result in very low ratings in terms of weight and dimensions metrics, which were used to evaluate profitability. Another implication of this design is that it is not compatible with the layout of Stryke due to the width of the touchboard exceeding the width of the indoor range (which is 1.5 m, referring to Section 2.2.2). It should also be noted that the pad is not designed with resisting knife strikes in mind therefore significant modification is required to make it compatible with quick draw.

8 Conclusion

Currently, there does not exist a commercially available or clear solution that satisfies the objective of Stryke for quick draw timing practice. The ones currently available either pose problems that would violate one or many constraints or are not designed for knife throwing. Based off the current engineering opportunity, stakeholders, engineering requirements, and data, the opportunity to develop a new and specialized system, namely the light and timing device (LTD) is justified. With the demand for a proposed device from our stakeholders, it is believed that the Stryke Target Range has an opportunity to develop a quick draw timing system which can be utilized for competitions and help improve the general user experience in knife throwing while encouraging new users to join the game.

9 Reference List

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[3] Li, L. (2018). *Set of images for RFP drawn on Sketch*.

[4] Kilfoyle, B. (n.d.). *THE CANADIAN NATIONAL KNIFE AND TOMAHAWK THROWING COMPETITION*.

[5] Dayaram, R. (2017). *Sneak peak of the Finger lakes Knife throw comp*.

[6] Li, L. (2018). *Images of Stryke taken during visits.*

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[8] En.wikipedia.org. (n.d.). *French cleat*. [online] Available at: https://en.wikipedia.org/wiki/French_cleat [Accessed 10 Feb. 2018].

[9] “Knife,” Wikipedia, 18-Feb-2018. [Online]. Available: https://en.wikipedia.org/wiki/Knife. [Accessed: 18-Feb-2018].

[10] C. Edwards, “Direct Sunlight Damage to an LCD TV Screen,” Techwalla. [Online]. Available: https://www.techwalla.com/articles/direct-sunlight-damage-to-an-lcd-tv-screen. [Accessed: 18-Feb-2018].

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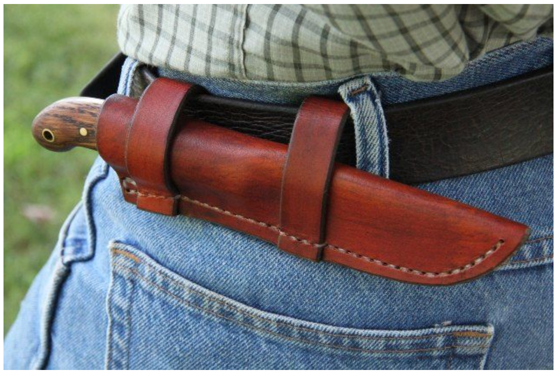
Appendix A: Source Extracts

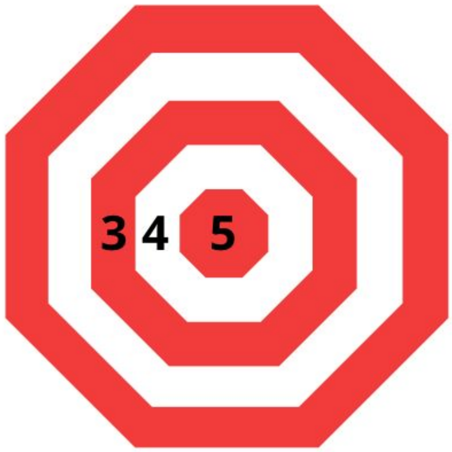
[1] "Frontiersman Games", *International Knife Throwers Hall of Fame (IKTHOF)*. [Online]. Available: http://ikthof.com/about/. [Accessed: 03- Feb- 2018].

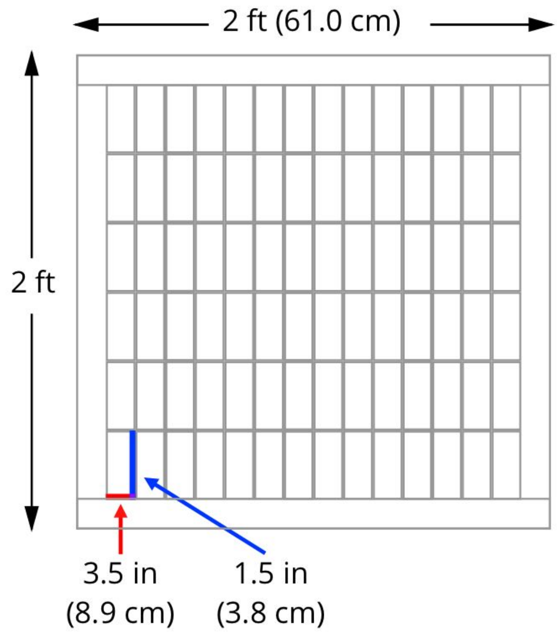
> **“Fast Draw**
>
> A Mountain Man knife at least 12 inches long and a minimum of 12 ounces in weight with a leather, wood, or bone handle will be used for both the Fast Draw and the Texas Three Step. Knife must be completely in the sheath: any part of the blade that is showing before being pulled will be warned once; the second time, the thrower will be disqualified. Minimum Distance 3 Meters knife must be drawn from the belt on the thrower’s waist. The SHEATH MUST be attached to the belt FIRMLY. The sheath may not be held higher than the belt by any means. To indicate readiness to compete, throwers must hold throwing arms stretched out towards the target, PALM UP (*EXECUTIVE DECISION*). *Must be in the 3 ring to score*. A maximum of three drops by BOTH the competitors. After three drops by BOTH competitors, BOTH throwers are disqualified. The individual competition winner will be BEST OUT OF THREE TRIES.”

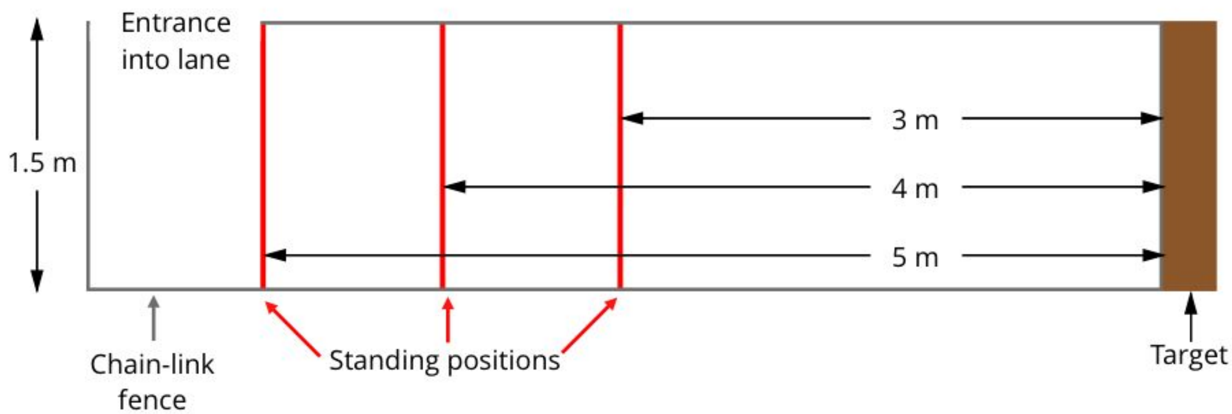
[2] Forge. L. (n.d.), *Jack Pine Jr. Deluxe Sheath on Belt*.

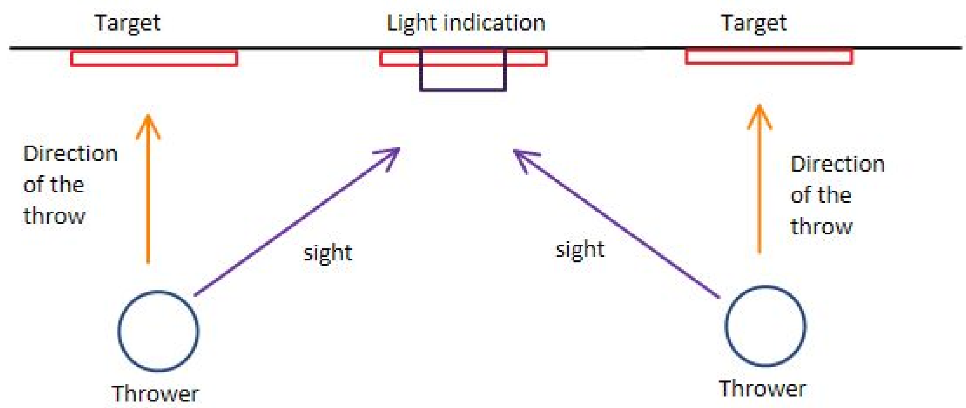


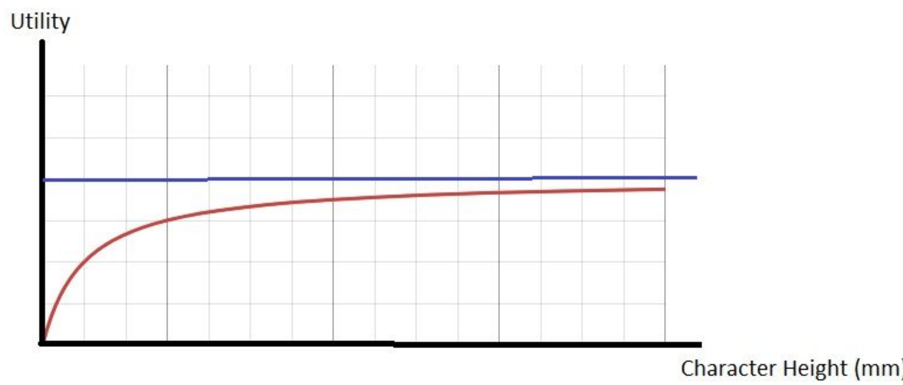
[3] Li, L. (2018). *Set of images for RFP drawn on Sketch*.


3
4
5


2 ft (61.0 cm)
2 ft
3.5 in
(8.9 cm)
1.5 in
(3.8 cm)


Entrance
into lane
1.5 m
3 m
4 m
5 m
Chain-link
fence
Standing positions
Target

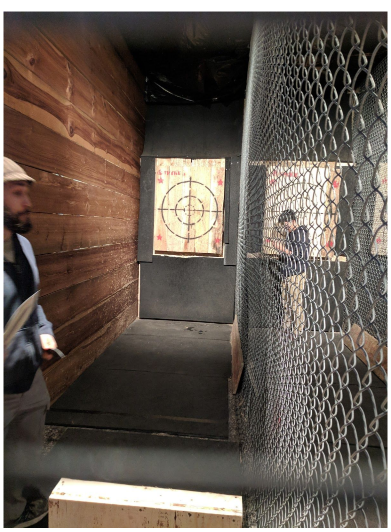




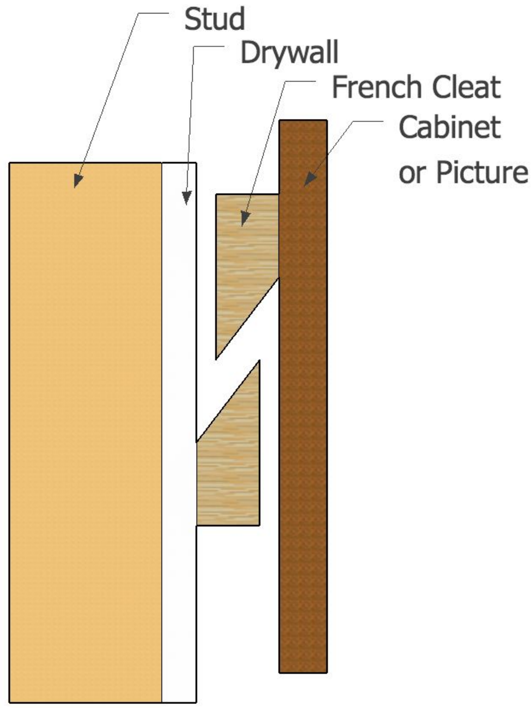
[4] Kilfoyle, B. (n.d.). *THE CANADIAN NATIONAL KNIFE AND TOMAHAWK THROWING COMPETITION*.

[5] Dayaram, R. (2017). *Sneak peak of the Finger lakes Knife throw comp*.

[6] Li, L. (2018). *Images of Stryke taken during visits.*



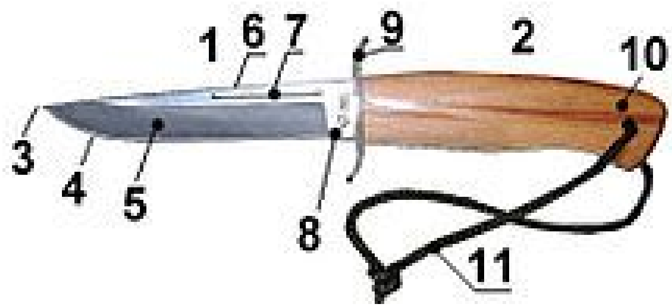
[7] J. Williams. (n.d.) *French Cleat Diagram.*



[8] En.wikipedia.org. (n.d.). *French cleat*. [online] Available at: https://en.wikipedia.org/wiki/French_cleat [Accessed 10 Feb. 2018].

> “A **French cleat** is way of securing a cabinet, mirror, artwork or other object to a wall. It is a molding with a 30–45 degree slope used to hang cabinets or other objects.
>
> French cleats can be used in pairs, or with a cleat mounted to the wall and a matching edge cut into the object to be hung.”

[9] "Knife," Wikipedia, 18-Feb-2018. [Online]. Available: https://en.wikipedia.org/wiki/Knife. [Accessed: 18-Feb-2018].



[10] C. Edwards, "Direct Sunlight Damage to an LCD TV Screen," Techwalla. [Online]. Available: https://www.techwalla.com/articles/direct-sunlight-damage-to-an-lcd-tv-screen. [Accessed: 18-Feb-2018].

> "**LCDs**
> LCDs use special materials that change their forms at a molecular level when an electrical current is applied to them. An LCD TV screen or monitor has a front panel made up of thousands of tiny dots of liquid crystal. The whole screen is backlit by a fluorescent panel. In a color screen, each pixel is made up of three liquid crystal dots, with each dot fronted by a color filter: one red, one blue and one green filter for each pixel. The liquid crystal dots can be made transparent or opaque depending on which colors are needed to make the specific color of a pixel.
>
> **Heat Damage**
>
> Direct sunlight falling onto an LCD screen can cause it to become very hot. Just as an LCD TV could be damaged if you installed it near a radiator or another strong heat source, direct sunlight can also affect your TV. The screen's plastic enclosure can be deformed by heat. The liquid crystal in the LCD screen can break down, meaning the display won't work. Other components in the LCD screen can malfunction as they aren't designed to tolerate high heat.
>
> **Ultraviolet Light**

Sunlight contains strong UV rays -- the same rays that cause sunburn. These UV rays can damage the long chain organic molecules that make up liquid crystals, causing the chains of molecules to break down. The tiny dots making up the screen no longer respond to the electric current by becoming completely transparent or completely opaque, meaning that light will pass through when it should not and light that should be allowed to pass will not. The damage is cumulative; over time, an LCD TV exposed to strong UV rays will lose definition, contrast and brightness."

[11] J. Lucas, "What Is Ultraviolet Light?," LiveScience, 15-Sep-2017. [Online]. Available: https://www.livescience.com/50326-what-is-ultraviolet-light.html. [Accessed: 18-Feb-2018].

"**UV effects**

Most of the natural UV light people encounter comes from the sun. However, only about 10 percent of sunlight is UV, and only about one-third of this penetrates the atmosphere to reach the ground, according to the National Toxicology Program (NTP). Of the solar UV energy that reaches the equator, 95 percent is UVA and 5 percent is UVB. No measurable UVC from solar radiation reaches the Earth's surface, because ozone, molecular oxygen and water vapor in the upper atmosphere completely absorb the shortest UV wavelengths. Still, "broad-spectrum ultraviolet radiation [UVA and UVB] is the strongest and most damaging to living things," according to the NTP's "13th Report on Carcinogens."

[12] T. Roeder, *Managing Project Stakeholders: Building a Foundation to Achieve Project Goals*. Hoboken: Wiley, 2013.

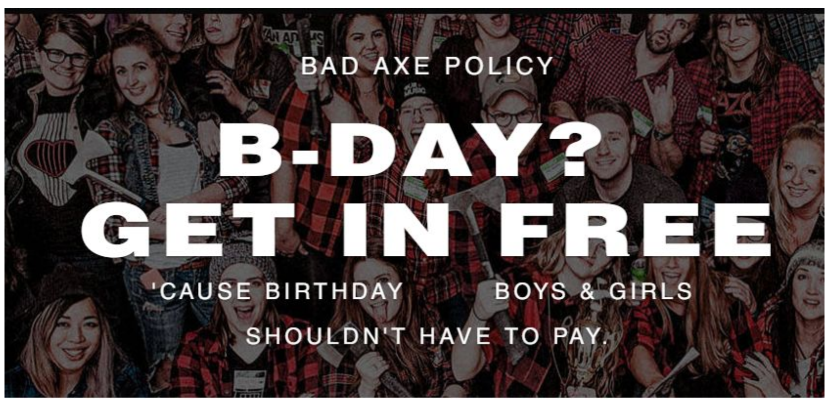
"**Power**
Power is the measure of a stakeholder's level of authority. Level of authority refers to the stakeholder's overall organizational power. Senior executives, by this definition, are likely to have significant power. Power, however, is not a measure of the stakeholder's ability to affect the particular project. There is a different word for this: impact.

Impact
Impact is a stakeholder's ability to effect changes to the project's planning or execution. A stakeholder with high impact can direct the project in their desired direction. It is possible for a stakeholder to have low power but high impact, and vice versa. See the inset for more detail.

Interest
Interest is a stakeholder’s level of concern regarding the project outcomes. A stakeholder with high interest is very concerned about the project. A stakeholder with low interest, however, has other priorities that are more important than the outcomes of this particular project.

Influence
Influence is the stakeholder’s level of active involvement in the project. A stakeholder with a high level of influence will participate more frequently in team discussions, meetings, and so on. A stakeholder with a low level of influence is not likely to participate in most team meetings.”

[13] “Bad Axe Toronto”. Bad Axe Throwing Inc. n.d. [Online] Available:https://badaxethrowing.com/locations/axe-throwing-toronto.[Accessed 04-Feb-2018]



Free birthday policy for kids at Bad Axe Throwing.

[14] M. Bainton, "Knife Throwing and Your Health", *International Knife Throwers Hall of Fame (IKTHOF)*, 2015. [Online]. Available: http://ikthof.com/wp-content/uploads/2015/09/Knife-Throwing-and-Your-Health.pdf. [Accessed: 04- Feb- 2018].

“Eccentric Isotonic exercise, such as knife throwing entails, is also called 'active range of motion exercise.' The muscle shortens during Isotonic exercise and muscle contraction causes a rise in the heart rate and a marked increase in stroke volume that results in an increase in cardiac output and a net decrease in peripheral resistance (due to vasodilation of contracted muscles). Isotonic exercise such as knife throwing will produce a moderate rise in systolic blood pressure, but the diastolic pressure usually remains unchanged. Anaerobic exercise is typically used by athletes in non-endurance sports to build power. Muscles that are trained under anaerobic conditions develop biologically differently giving them greater performance in short duration activities. The contraction and release of muscles from start to finish of a single knife throw is a good example of this. Knife throwing utilizes the entire muscular structure in the act of throwing a knife."

[15] "About", *International Knife Throwers Hall of Fame (IKTHOF)*. [Online]. Available: http://ikthof.com/about/. [Accessed: 03- Feb- 2018].

"The Mission of the IKTHOF is: To Honor, Preserve, Educate and Promote:

To honor individuals who have made outstanding contributions to knife and tomahawk throwing, knife making, and education of the public in support of knife throwing.

To preserve professional knife and tomahawk throwing historic documents and artifacts.

To educate the public regarding the origin, development, and growth of knife and tomahawk throwing as an important part of the American culture.
To promote the positive values of the sport."

List of international competition winners (held in Ontario Canada for 2017). Shows the variety of different types of competitions that vary in accuracy, speed and distance won by certain individuals. Note that this is not an exhaustive list of all the competitions held by IKTHOF.

2017 Ontario, Canada

World Champion Gold Cup Knife – Pete Baldwin – Ontario – Canada
World Champion Gold Cup Hawk – Dave " Chef " Cox – USA
World Champion Knife Thrower Pro – Richard " Bulls Eye" – Wesson USA
Word Champion Knife Thrower No Spin – Neil Kehler – Alberta – Canada
World Champion Hawk Thrower – Pro – Pete Baldwin – Ontario – Canada
World Champion Silhouette – Richard "Bulls Eye " Wesson – USA
World Champion Fast Draw – Paul Maccarone – New York – USA
World Champion Speed Throw – Bill Nichols – 25 – Florida – USA
World Champion Texas Three Step – Brett Dewhirst – Ontario – Canada
World Champion Long Distance Hawk – David Payne – Ontario – Canada 74' 2"
World Champion Long Distance Knife – Scott Macrae – Ontario Canada 48' 8"

[16] Thomas Eye Group. (n.d.). *What Is Normal Vision Atlanta | What Isn't Normal Vision Atlanta*. [online] Available at: https://www.thomaseye.com/normal-vision-atlanta.htm [Accessed 4 Feb. 2018].

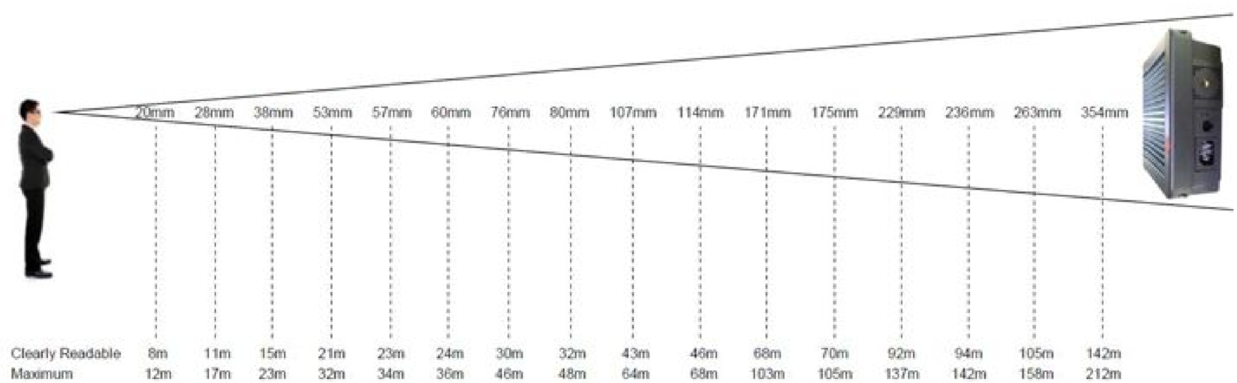
> "**What Is Normal Vision?**
>
> Studies have found that "normal" vision is being able to see a certain size line on the eye chart (the Snellen Chart) from 20 feet away. If you have 20/20 vision, you can see clearly at 20 feet what should normally be seen at that distance. If you have 20/40 vision, it means that you must be as close as 20 feet to see what a person with normal vision can see at 40 feet. So, the first number refers to the distance you stand from the chart, and the second number is the distance a person with normal vision could read the same line you correctly read – the larger the second number, the worse the vision.
>
> 20/20 does not necessarily mean perfect vision. Some people can see well at a distance, but are unable to bring closer objects into focus because of hyperopia (farsightedness) or presbyopia (loss of focusing ability). In order to diagnose any causes that may be affecting your ability to see well, we recommend scheduling a comprehensive eye examination at one of our Atlanta area Thomas Eye Group locations including Lilburn, Lithonia, Newnan, Roswell, Sandy Springs, Suwanee and Woodstock."

[17] T. Galassi, "UNITED STATES DEPARTMENT OF LABOR," Occupational Safety and Health Administration, 05-Apr-2015. [Online]. Available: https://www.osha.gov/pls/oshaweb/owadisp.show_document?p_table=INTERPRETATIONS&p_id=29936. [Accessed: 18-Feb-2018].

> "NIOSH has a lifting equation (discussed in the above-referenced *Applications Manual*) for calculating a *recommended weight limit* for one person under different conditions. The lifting equation establishes a maximum load of 51 pounds, which is then adjusted to account for how often you are lifting, twisting of your back during lifting, the vertical distance the load is lifted, the distance of the load from your body, the distance you move while lifting the load, and how easy it is to hold onto the load."

[18] Monsters Edge on behalf of Speckled Frog, “Text Height & Viewing Distance,” Reception Displays, Lobby Displays, Viewing Distance | Spectra Displays Ltd. [Online]. Available: http://spectra-displays.co.uk/help-and-support/technical-info/text-height-and-viewing-distance. [Accessed: 18-Feb-2018].

> “Viewing distance is critical when deciding on a display and this simple formula helps determine the best size of characters.
> - Character height in mm multiplied by 0.4 gives the clear readable viewing distance in metres.
> - Character height in mm multiplied by 0.6 gives the maximum viewing distance in metres.”



[19] “Recommended Light Levels,” Recommended Light Levels (Illuminance) for Outdoor and Indoor Venues. *https://www.noao.edu/education/QLTkit/ACTIVITY_Documents/Safety/LightLevels_outdoor+indoor.pdf*

Measuring Units of Light Level - Illuminance

Illuminance is measured in foot candles *(ftcd, fc, fcd)* or lux (in the metric SI system). A *foot candle* is actually *one lumen* of *light density per square foot*; *one lux* is *one lumen per square meter*.

- *1 lux = 1 lumen / sq meter = 0.0001 phot = 0.0929 foot candle (ftcd, fcd)*
- *1 phot = 1 lumen / sq centimeter = 10000 lumens / sq meter = 10000 lux*
- *1 foot candle (ftcd, fcd) = 1 lumen / sq ft = 10.752 lux*

Common Light Levels Outdoors from Natural Sources

Common light levels outdoor at day and night can be found in the table below:

Condition	**Illumination**	
	(ftcd)	***(lux)***
Sunlight	10,000	107,527
Full Daylight	1,000	10,752
Overcast Day	100	1,075
Very Dark Day	10	107
Twilight	1	10.8
Deep Twilight	.1	1.08
Full Moon	.01	.108
Quarter Moon	.001	.0108
Starlight	.0001	.0011
Overcast Night	.00001	.0001

Common Light Levels Outdoors from Manufactured Sources

The nomenclature for most of the types of areas listed in the table below can be found in the City of Los Angeles, Department of Public Works, Bureau of Street Lighting’s “DESIGN STANDARDS AND GUIDELINES” at the URL address under **References** at the end of this document.

[20] C. Roseberry, “What Are Portable Devices?,” *Lifewire*. [Online]. Available: https://www.lifewire.com/what-are-portable-devices-2377121. [Accessed: 18-Feb-2018].

“**Portable Devices**

There's no standard definition for "portable device," although this term probably has been in use longer than the term "mobile device." As the name implies, a portable device simply means something that is small and lightweight enough to move around and carry with relative ease."

[21] M. L. Wang and M. R. Luo, "Effects of LED lighting on office work performance," *2016 13th China International Forum on Solid State Lighting (SSLChina)*, Nov. 2016.

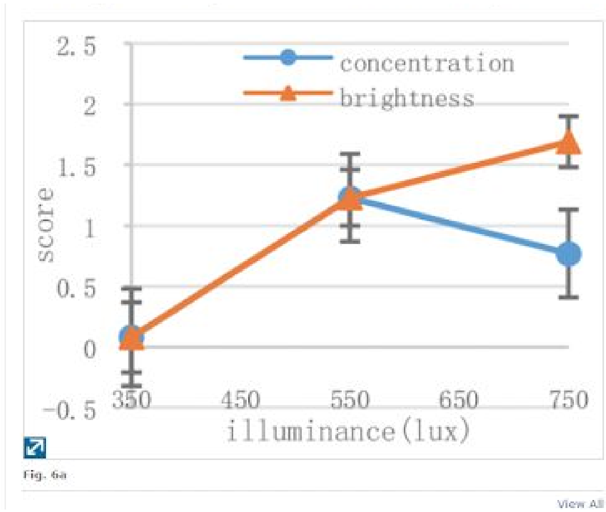
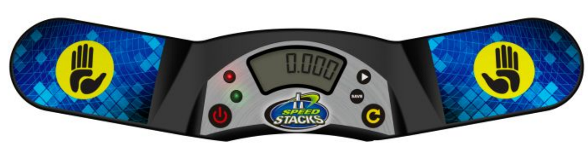
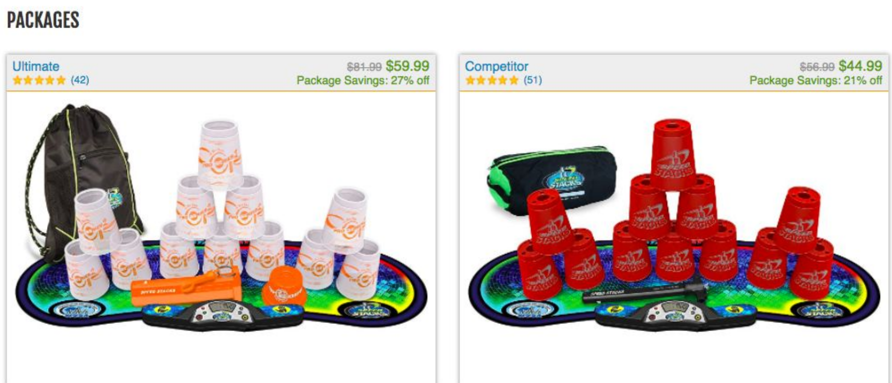


Figure X: In the first graph it can be seen that the optimum comfort level is at midway.

[22] "Speed Stacks Store (Sport Stacking)", *SpeedStacks.com*. [Online]. Available: https://www.speedstacks.com/store/retail/. [Accessed: 18- Feb- 2018].





[23] R. Fox and J. Goers, "Mat for timing competitions", US6940783B2, 2002.

> "The sport involves stacking and unstacking a set of specially designed cups in pre-arranged sequences while being timed. The object of the competitions is to complete the sequence or sequences in as short a time as possible. There are several standard sequences and the competitions can be performed by an individual or by a team in a relay fashion.
>
> Timing of the competition is usually performed by a judge with a stopwatch. The competitor begins with both hands face down on the table where the cups are to be stacked. The judge gives a verbal cue, usually "Ready, Get Set, Go", starting the stopwatch on the word GO. Time is stopped when the last cup is down stacked in the particular sequence.
>
> One of the major drawbacks to the sport has been the inaccuracy associated with having different individuals time the competitors using stopwatches. Errors in timing due to variations in human reaction time often exceed several tenths of a second and are significant (as much as 10% off) when measuring competitive times. Competitive times in this sport range from 2 to 15 seconds with winners of the competition usually determined by hundredths of seconds. To mitigate against the human reaction time, in final competitions, three judges are used and the high and low times are not counted against the competitor. This measure, however, is

inefficient and does not cure the inaccuracies inherent with using human judges since the measured time to complete the sequence is still subjective."

Another embodiment of the present invention describes an apparatus comprising a mat, a pressure sensitive trigger connected to the mat, and a timer operatively connected to the trigger so that the timer begins to accrue time with a first activation of the trigger and stops with a second activation of the trigger."

The timer mechanism can be any kind of timer that meets the following criteria: 1) is capable of timing to at least 0.01 seconds; 2) able to be started and stopped by the competitor without reliance on any third party; 3) predictable, repeatable, reusable, and reliable; 4) does not interfere with the stacking process; and 5) able to be operated using a free standing power source, such as batteries, for long periods of time without resort to auxiliary power sources. A timer found suitable for use with the present invention includes a timer constructed from standard electronic components, including a microprocessor to accurately keep time, an LED display, discrete light emitting diodes, a reset switch, a power switch and a power source."

[24] G. Elert, “Speed of a Bullet,” Speed of a Bullet - The Physics Factbook, 1999. [Online]. Available: https://hypertextbook.com/facts/1999/MariaPereyra.shtml. [Accessed: 18-Feb-2018].

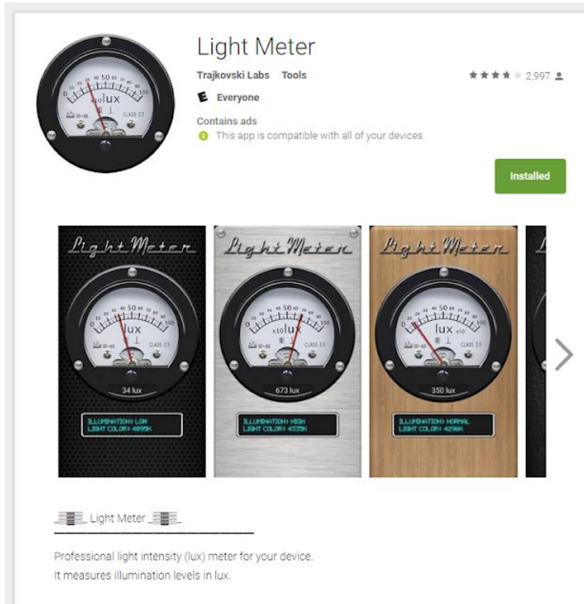
Bibliographic Entry	Result (w/surrounding text)	Standardized Result
Cutnell, John D. & Johnson, Kenneth W. *Physics.3rd ed*. New York: Wiley, 208-209.	"With this value for v_f, it is now possible to determine the speed of the bullet … +896 m/s"	896 m/s
Ballistics. *The World Book Encyclopedia*. New York: World Book, 1998.	"With modern propulsion techniques, the projectile's initial velocity may be as high as 4000 feet (1200 meters) per second for some rifles and 5000 feet (1500 meters) per second for some large guns."	1200–1500 m/s
Bullet. *The World Book Encyclopedia*. New York: World Book, 1998.	"The *velocity* (speed) of rifle bullets varies between 600 and 5000 feet (180 and 1500 meters) per second. Some bullets can hit targets as far away as 6000 yards (5000 meters)."	180–1500 m/s
Petzal, David E. "How fast is a speeding bullet." *Field and Stream*. 97 (1992): 23.	.22 rimfire cartridge 1200–1500 fps .22 centerfire cartridge 2400–3000 fps .22 Swift 4000 fps .38 Special 600 fps .221 Fireball 2650 fps	180–1220 m/s

[25] C. Thiel, "Physics of knife throwing", *Knifethrowing.info*. [Online]. Available: http://www.knifethrowing.info/physics_of_knife_throwing.html. [Accessed: 06- Feb- 2018].

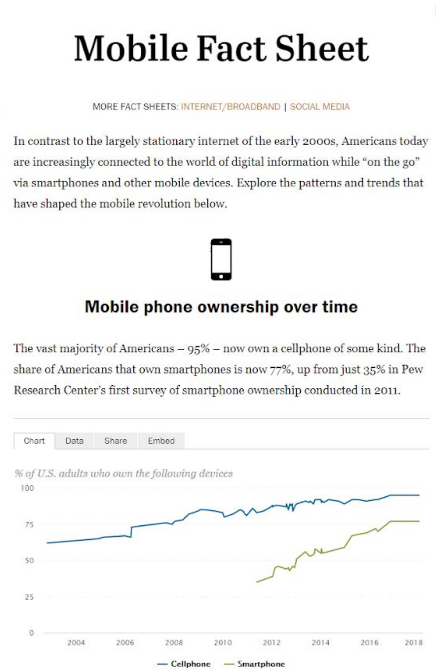
Initial speed [km/h]	Angle of throw [degree]	Real distance on arch [m]
35,66	45,00	11,47
50,43	15,00	10,12
75,18	6,50	10,02
99,09	3,72	10,01

Estimate a knife throw to be between 6.5 and 15.00 degrees; 60 km/h.

[26] “Light Meter - Android Apps on Google Play,” *Google*. [Online]. Available: https://play.google.com/store/apps/details?id=com.bti.lightMeter. [Accessed: 18-Feb-2018].



[27] “Mobile Fact Sheet,” Pew Research Center: Internet, Science & Tech, 05-Feb-2018. [Online]. Available: http://www.pewinternet.org/fact-sheet/mobile/. [Accessed: 18-Feb-2018].



[28] "How It Works – Target Impact Signal System", *Targetimpactsignal.com*. [Online]. Available: http://targetimpactsignal.com/how-it-works/. [Accessed: 12- Feb- 2018].

> “Fasten the TISS sensor to the back of an impenetrable metal target (industrial mounting strips included). The sensor signals the strobe light to flash each time a bullet strikes the target. The target impact sensor is connected to the flash unit with a 10 to 20 foot cable,, which is included in the system.
>
> - Flash is visible at 1,000 meters (1093 yards) or greater with the naked eye.”

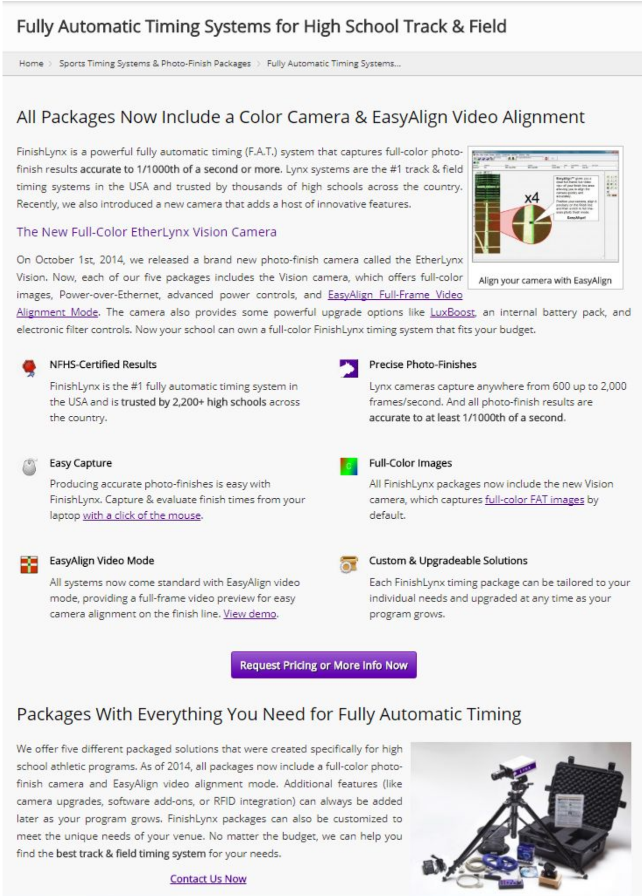
[29] “Brightness vs. Distance”. Berkeley. [Online]. Available:
http://bccp.berkeley.edu/o/Academy/workshop08/08%20PDFs/Inv_Square_Law.pdf
[Accessed: 12- Feb- 2018].

Brightness

- Definition for a STAR: the amount of energy that lands on a square meter of Earth every second.
- Unit: watts/square meter (W/m^2)
- Similar for a light bulb: The unit is lumen/sq. meter which is called a LUX

[30] *Using the TISS I at 50 Yards*. Youtube, 2017.

[31] “Track & Field Timing Systems for High School - Packages,” *FinishLynx*. [Online]. Available: http://www.finishlynx.com/packages/high-school/. [Accessed: 18-Feb-2018].



Fully Automatic Timing Systems for High School Track & Field

Home › Sports Timing Systems & Photo-Finish Packages › Fully Automatic Timing Systems...

All Packages Now Include a Color Camera & EasyAlign Video Alignment

FinishLynx is a powerful fully automatic timing (F.A.T.) system that captures full-color photo-finish results **accurate to 1/1000th of a second or more.** Lynx systems are the #1 track & field timing systems in the USA and trusted by thousands of high schools across the country. Recently, we also introduced a new camera that adds a host of innovative features.

Align your camera with EasyAlign

The New Full-Color EtherLynx Vision Camera

On October 1st, 2014, we released a brand new photo-finish camera called the EtherLynx Vision. Now, each of our five packages includes the Vision camera, which offers full-color images, Power-over-Ethernet, advanced power controls, and EasyAlign Full-Frame Video Alignment Mode. The camera also provides some powerful upgrade options like LuxBoost, an internal battery pack, and electronic filter controls. Now your school can own a full-color FinishLynx timing system that fits your budget.

NFHS-Certified Results
FinishLynx is the #1 fully automatic timing system in the USA and is **trusted by 2,200+ high schools** across the country.

Precise Photo-Finishes
Lynx cameras capture anywhere from 600 up to 2,000 frames/second. And all photo-finish results are **accurate to at least 1/1000th of a second.**

Easy Capture
Producing accurate photo-finishes is easy with FinishLynx. Capture & evaluate finish times from your laptop with a click of the mouse.

Full-Color Images
All FinishLynx packages now include the new Vision camera, which captures full-color FAT images by default.

EasyAlign Video Mode
All systems now come standard with EasyAlign video mode, providing a full-frame video preview for easy camera alignment on the finish line. View demo.

Custom & Upgradeable Solutions
Each FinishLynx timing package can be tailored to your individual needs and upgraded at any time as your program grows.

Request Pricing or More Info Now

Packages With Everything You Need for Fully Automatic Timing

We offer five different packaged solutions that were created specifically for high school athletic programs. As of 2014, all packages now include a full-color photo-finish camera and EasyAlign video alignment mode. Additional features (like camera upgrades, software add-ons, or RFID integration) can always be added later as your program grows. FinishLynx packages can also be customized to meet the unique needs of your venue. No matter the budget, we can help you find the **best track & field timing system** for your needs.

Contact Us Now

[32] G. Pitcher, Decreasing Dimensions, vol. 37. Elsevier Inc.

Abstract: The design solutions offered by digital technology from Seagate for **portable** consumer electronics are discussed. Seagate has launched a 1in drive in the shape of the ST1 range that can store upto 5Gbyte. The challenge underlying this is to put **smaller** heads on **smaller** motors and applying **smaller** electronic packaging. Seagate continues to work on new compounds for recording heads, looking for the most magnetically stable and hard wearing.

Author Pitcher Graham discusses portability and how the equivalent to a portable design is one which is smaller with smaller components.

[33] “Compare the Packages .” [Online]. Available: http://www.finishlynx.com/download/marketing-documents/high-school-track/packages-compare.pdf. [Accessed: 18-Feb-2018].

Compare the Packages

Package Contents	Scholastic $4,995	Bronze $7,495	Silver $11,295	Gold $16,995	MVP $24,995
Upgradeable to a Higher Specification:	YES	YES	YES	YES	YES
Number of Photo-Finish Cameras:	1	1	1	1	2
o Primary EtherLynx Camera Model	5L510	5L500 Vision	5L500 Vision	5L500 Vision	5L500 Vision
o Standard Frame Rate (frames per sec.)	600	1,000	2,000	2,000	2,000
o Lens	6mm Fixed	Manual Zoom	Motorized Zoom	Motorized Zoom	Motorized Zoom
o **EasyAlign** – Full-Frame Video Alignment	✓	✓	✓	✓	✓
o **Full-Color** Photo-Finish Images	✓	✓	✓	✓	✓
o 500’ Start Cable & Cabled Start Sensor	✓	✓	✓	✓	✓
o All necessary Ethernet & Camera Cables	✓	✓	✓	✓	✓
o Water Resistant Housing	✓	✓	✓	✓	✓
o Power-over-Ethernet Injector	✓	✓	✓	✓*	✓*
o Custom FinishLynx Carrying Case	✓	✓	✓	✓	✓
o LynxPad Meet Management Software	✓	✓	✓	✓	✓
o Tech Support & 1-Year Renewable Warranty	✓	✓	✓	✓	✓
o Camera Tripod(s) and Mounting Hardware	✓	✓	✓	✓	✓
o **FinishLynx Software**	LITE	FULL	FULL	FULL	FULL
~ Support for RadioLynx Wireless Start	✓	✓	✓	✓	✓
~ Support for Lynx Software Plug-in(s)		✓	✓	✓	✓
~ Support for Multiple FinishLynx Cameras		✓	✓	✓	✓
~ Support for IdentiLynx 2-D Video Camera		✓	✓	✓	✓
~ Scoreboard Interface for Display of Running Time & Results		✓	✓	✓	✓
~ Wind Gauge Interface for Automatic Wind Reading Data Import		✓	✓	✓	✓
o Automatic Capture & Virtual Photocell Plug-in			✓	✓	✓
o High Resolution (2,000 fps) Images			✓	✓	✓
o Remote Positioner (Remote Camera Adjustment)			✓	✓	✓
o Motorized Lens (Remote Lens Adjustment)			✓	✓	✓
o RadioLynx Wireless Start System				✓	✓
o *Power-over-Ethernet Network Switch				✓	✓
o IdentiLynx Full-Frame Video Camera				✓	✓
o 9-Digit Alphanumeric Scoreboard				✓	✓
o IAAF Approved Sonic Wind Gauge					✓
o Secondary EtherLynx Camera Model (Reverse Angle / Alternate Finish Line)					5L500 Vision
~ Standard Frame Rate (frames per sec.)					*1,000*
~ Remote Positioner (Remote Camera Adjustment)					✓
~ Motorized Lens (Remote Lens Adjustment)					✓
~ Upgradeable to a Higher Specification					✓

[34] “Touchpads Daktronics,” *Aquam’s Institutional catalog - Canada*. [Online]. Available: http://www.aquam.com/2/Catalogs/Institutional/Touchpads Daktronics.php. [Accessed: 18-Feb-2018].

TOUCHPADS DAKTRONICS

Daktronics touchpads use a proven design that incorporates three conductive plates made of noncorrosive stainless steel. The stainless-steel construction gives years of reliable service and will not crack or break under pressure. The front and top of the touchpad provide high sensitivity across the entire non-slip surface of the touchpad so that competitive swimmers can rely on fair and consistent starts and turns. These touchpads fill up with water to equalize the internal and external water pressure, making leaking and floating problems nonexistent. They are also designed to be self-draining, so that water quickly drains through small holes along the bottom.

[35] “T-7000 SERIES TOUCHPADS PRODUCT SPECIFICATIONS.” [Online]. Available: https://www.daktronics.com/Web%20Documents/HSPR-Documents/SL10421.pdf. [Accessed: 18-Feb-2018].

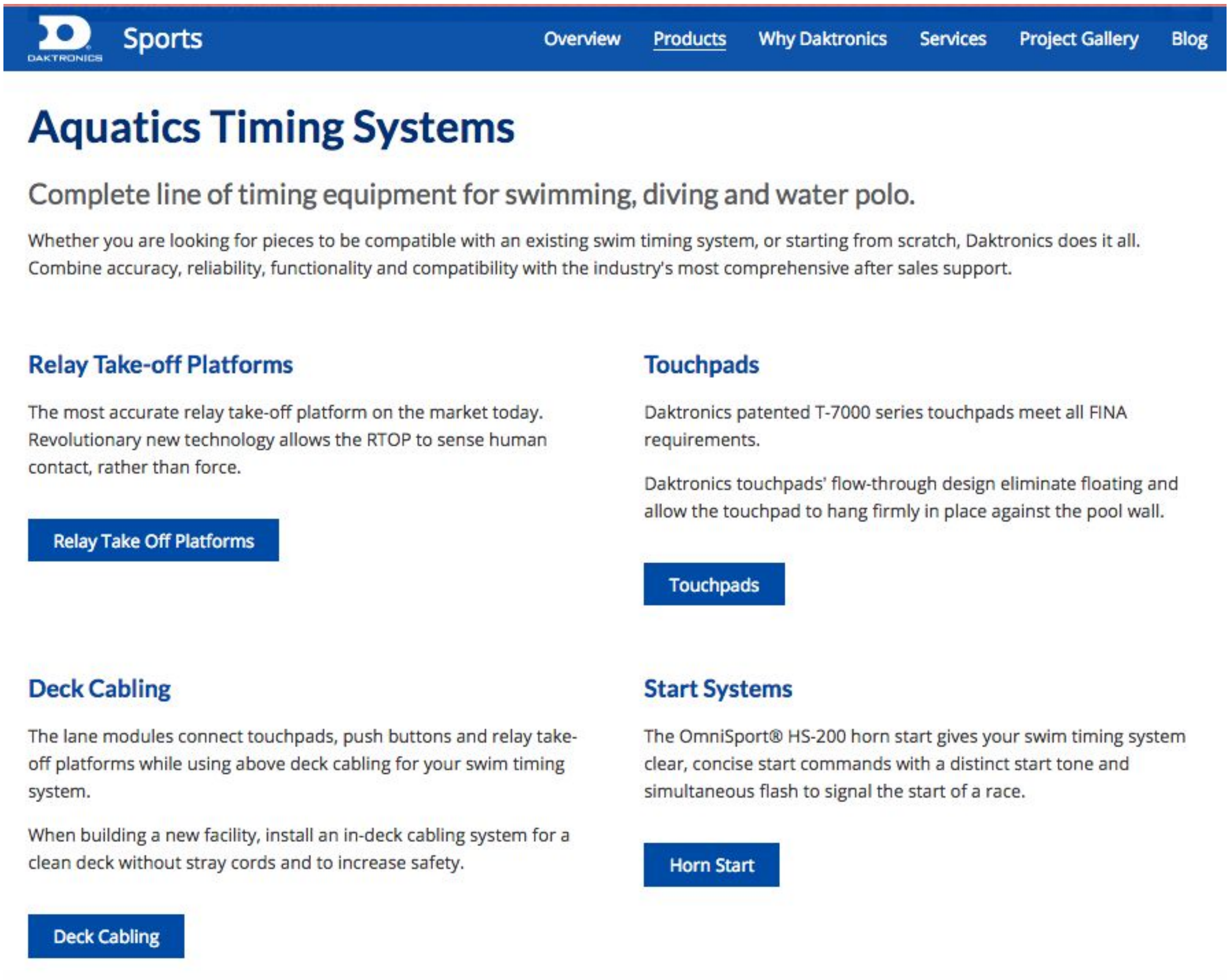
DAKTRONICS

T-7000 SERIES TOUCHPADS PRODUCT SPECIFICATIONS

Model	Dimensions *Height, Width, Depth (of top flange)*	Weight
T-7060	H 1'-10", W 5'-0", D 2.8" (559 mm, 1524 mm, 71 mm)	28.25 lb (12.8 kg)
T-7078	H 1'-10", W 6'-6", D 2.8" (559 mm, 1981 mm, 71 mm)	37.5 lb (17 kg)
T-7096	H 2'-0", W 8'-0", D 2.8" (610 mm, 2438 mm, 71 mm)	50 lb (22.7 kg)
FT-7150	H 3'-0", W 5'-0", D 2" (914 mm, 1524 mm, 51 mm)	44.7 lb (20.3 kg)
FT-7190	H 3'-0", W 6'-3", D 2" (914 mm, 1905 mm, 51 mm)	56.8 lb (25.8 kg)
FT-7190T	H 3'-0", W 6'-3", D 3.9" (914 mm, 1905 mm, 99 mm)	57 lb (25.9 kg)
FT-7240T	H 3'-0", W 7'-11", D 3.9" (914 mm, 2413 mm, 99 mm)	72.5 lb (32.9 kg)

Note: All touchpad models have a thickness of 0.3" (8 mm).

[36] “Aquatics Timing Systems.” [Online]. Available: https://www.daktronics.com/en-us/products/sports/aquatics/swim-timing-systems. [Accessed: 18-Feb-2018].



Sports

Overview Products Why Daktronics Services Project Gallery Blog

Aquatics Timing Systems

Complete line of timing equipment for swimming, diving and water polo.

Whether you are looking for pieces to be compatible with an existing swim timing system, or starting from scratch, Daktronics does it all. Combine accuracy, reliability, functionality and compatibility with the industry's most comprehensive after sales support.

Relay Take-off Platforms

The most accurate relay take-off platform on the market today. Revolutionary new technology allows the RTOP to sense human contact, rather than force.

Relay Take Off Platforms

Touchpads

Daktronics patented T-7000 series touchpads meet all FINA requirements.

Daktronics touchpads' flow-through design eliminate floating and allow the touchpad to hang firmly in place against the pool wall.

Touchpads

Deck Cabling

The lane modules connect touchpads, push buttons and relay take-off platforms while using above deck cabling for your swim timing system.

When building a new facility, install an in-deck cabling system for a clean deck without stray cords and to increase safety.

Deck Cabling

Start Systems

The OmniSport® HS-200 horn start gives your swim timing system clear, concise start commands with a distinct start tone and simultaneous flash to signal the start of a race.

Horn Start

[37] “FINA FACILITIES RULES.” [Online]. Available: https://www.fina.org/sites/default/files/finafacilities_rules.pdf. [Accessed: 18-Feb-2018].

FR 4 AUTOMATIC OFFICIATING EQUIPMENT

FR 4.1 Automatic and Semi-Automatic Officiating Equipment records the elapsed time of each swimmer and determines the relative place in a race. Judging and timing shall be to 2 decimal places (1/100 of a second). Equipment that is installed shall not interfere with the swimmers' starts, turns, or the function of the overflow system.

FR 4.2 The Equipment must:

FR 4.2.1 Be activated by the starter.

FR 4.2.2 Have no exposed wires on the pool deck, if possible.

FR 4.2.3 Be able to display all recorded information for each lane by place and by lane.

FR 4.2.4 Provide easy digital reading of a swimmer's time.

FR 4.3 Starting devices

FR 4.3.1 The starter shall have a microphone for oral commands.

FR 4.3.2 If a pistol is used, it shall be used with a transducer.

FR 4.3.3 Both the microphone and the transducer shall be connected to loudspeakers at each starting block where both the starter's commands and the starting signal can be heard equally and simultaneously by each swimmer.

FR 4.4 Touch panels for Automatic Equipment

FR 4.4.1 The minimum measurement of the touch panels shall be 2.4 metres wide and 0.9 metre high, and their thickness shall be 0.01 metre ± 0.002 metre. They shall extend 0.3 metre above and 0.6 metre below the surface of the water. The equipment in each lane shall be connected independently, so it may be controlled individually. The surface of the panels shall be of a bright colour and shall bear the line markings approved for the end walls.

FR 4.4.2 Installation - The touch panels shall be installed in a fixed position in the centre of the lanes. The panels may be portable, allowing the pool operator to remove them when there are no competitors.

FR 4.4.3 Sensitivity - The sensitivity of the panels shall be such that they cannot be activated by water turbulence, but will be activated by a light hand touch. The panels shall be sensitive on the top edge.

FR 4.4.4 Markings - The markings on the panels shall conform with and superimpose on the existing markings of the pool. The perimeter and edges of the panels shall be defined by a 0.025 metre black border.

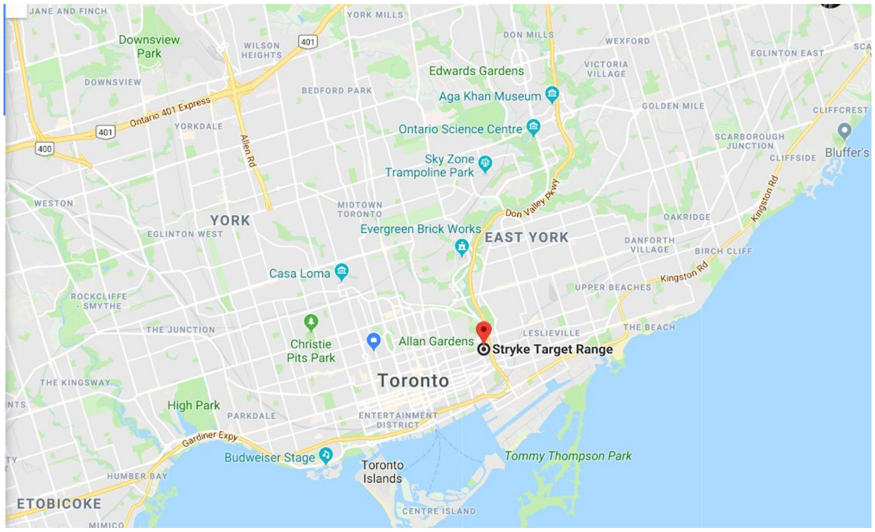
FR 4.4.5 Safety - The panels shall be safe from the possibility of electrical shock and shall not have sharp edges.

FR 4.5 With Semi-Automatic Equipment, the finish shall be recorded by buttons pushed by timekeepers at the finish touch of the swimmer.

Appendix B: Field Notes

1 Location and Contact Info

Address: Stryke Target Range, 738 Dundas St E, Toronto, ON M5A 2C3



Ryan Dayaram, Manager of Toronto Stryke location
Phone Number: 647-290-9391
Email: ryan@stryke.ca

Humphrey Tan, Stryke Employee
Email: Humprey@techie.com (PREFERRED METHOD OF CONTACT)

2 Transcript of Interview #1 with employee, Humphrey

Feb 1, 2018 at Stryke.

Humphrey: "So the object of the quick draw is to draw and throw to the target faster than your opponent. But there is kind of a caveat in this one. You actually have to hit it within a specific space. These aren't actually the targets that they use. They actually use targets more similar to the ones at the end."

Student 1: "Like a bullseye?"

Humphrey: "Yeah. But here's the thing, in competition they'll normally have a three box set up like this but they'll only use the center box for competition. And you have to score within this space. So 5, 4, and 3, are the point values and everything within the 3 box is the section you're looking for. Now to make this a little easier typically in competition these boxes are 2 by 2 squared out boxes. They are set up like this. So typically it's the center box we need. The other two don't really matter. They are typically mounted on a ___ board. So the light system will have to go somewhere up here. There's normally space above. It has to be decently rugged cause knives are being thrown at it. How this works is your 2 throwers would set up in the lane. So what'll happen is they both set up, arm out at shoulder length, ready, and then basically what'll happen is that the lighting system will start that counter and will automatically fire to green. Once it goes to green, the objective for the thrower is to draw and throw as quickly as possible. The trick now is the contact to the board. There's gonna be a flaw in this because if someone hits the board but it doesn't stick they don't actually win but the hardest part is that when you get to the higher levels a lot of the throwers are actually throwing at the same time so you'll hear a fraction of a hit difference but you won't know who actually hit it so this is where the timer comes in handy because it will actually sense that hit on that board the fraction of the second faster than the other so you'll know who the winner is assuming they both hit inside the target area."

Student 1: "So the only indicator for the thrower is the light on top that says throw. Where would you see the time difference?"

Humphrey: "In the mobile versions [not cellphone they mean portable] you'll see on that small unit there's a small screen on the bottom. I wouldn't mind having a larger LED read out over each target. So built into the lighting system could be that timer."

Student 1: "So you would want an accelerometer on the back of the board? So would you have to find an accelerometer to match the size of the board or would you make a board to match the size of an accelerometer?"

Humphrey : "So it doesn't have to be the entire board since each lane would be separate boards it would just have to pick up any kind of contact regardless of where it is on the board. I would have to pick up contact through the board from the other side."

Student 1: "Do you have to recognize if it actually hits the board?"

Humphrey: "No i wouldn't worry about that I'd just focus on the contact on its own. Because getting a scoring system that looks inside of a specific area makes it more complicated."

Student 2: "You could just eyeball that."

Humphrey: "Yeah there's always gonna be a ref anyway who's calling the start anyways because they have the remote. So they're gonna be the one to call the winner anyway."

Student 1: “So the remote would just reset both timers and turn it on or off?”

Humphrey: “Yeah so they’re be a ‘clear’ and ‘go’ essentially. 2 button timer.”

Student 1: “Would you like to store the data if its possible”

Humphrey: “It’d be really cool to store the data. What I’ve experienced in competition is that a lot of the time between each throw they actually walk up to the little box and they normally have it high up and they have to peep into this little thing that have the timers for each side. How we did have another problem tho. They used an LCD screen. Our box was in the sun”

Student 1: “So you couldn’t see it?”

Humphrey: “What happened was they basically burned the crystal so you couldn’t see anything. Or it overheated the crystal so you couldn’t actually see the number it was just a blank screen.”

Student 1: “So are most knife throwing competitions outdoors?”

Humphrey: “They’re typically outdoors yes. I dont think theres too many indoor ones cause typically they have one set of competitions which is longest knife or longest tomahawk and in order to do that when you’re throwing 30-50 feet you need that arc space so a lot of the time we don’t have that ceiling height to throw it.”

Student 1: “So we have to make one item accessible for both indoors and outdoors?”

Humphrey: “More for outdoors because you can set it up for indoors easily.”

Humphrey: “So this is how it’s set up, there are three individual boxes along one wall. Now they are normally only using the centre boxes. Now at this particular competition since the boards were on one giant wall, attached to one, we couldn’t put the sensors on the two consecutive lanes, we had to skip a lane in the middle. Each lane is about 8ft wide, it’s 12ft between one target and the other. They had to look over their shoulders which is a huge disadvantage for anyone that isn’t left handed. If it is positioned over the board you can see better.”

Student 4: “Do you think the lighting would be more of a disadvantage?”

Humphrey: “We are actually trained to go on green and as long as it’s not flashing it shouldn’t be a problem”

Student 1: " Do you want the part to be dynamic or do you want it fixed?, for example if set up on the middle board do you want it to be able to move it back and forth somewhere else?"

Humphrey: "I would want it as something more mobile, at the range it should be fixed but for the use in competition we want to be able to pack it all up, that's where indoor and outdoor should be able to occur."

Student 1: "How do you think you would attach the board to the accelerometer"

Humphrey: "In this circumstance, the boards were modular meaning that they were attached to the back of the boards"

Student 1: "Sir, can you show us how to remove the boards"

Humphrey: "So basically what is it, here, holly actually took picture of them, they're french cleats, so they actually just hook together" (referring to boards)

Student 1: "So do you remove this part to? (referring to the frame) or do you just remove this part? (referring to the 5 front wooden board pieces)?"

Humphrey: "When building the boards, we actually just remove these (referring to the 5 front wooden board pieces)'"

Student 1: "And you get to reuse this part? (referring to wooden frame)"

Humphrey: "Yeah, we would get to reuse the frame"

Student 1: "Do you like nail it in? or? (referring to the connection between the front 5 wooden pieces and the frame)"

Humphrey: "They're screwed"

Student 1: "Ok"

Humphrey: "They're screwed in the sense, so you can basically just unscrew them. Sometimes the screws break, so we have to snap them, because the heads have been hit (hit by knives/axes)."

Student 3: "So for the lights, ideally, you would want 1 set of lights above each set of targets?"

Humphrey: “Correct. I would be interested in seeing a lighting system with a timer or with an LED readout of time. The lighting system would basically be tri-color, red for loss, yellow for ready, and green for throw, and then, on the other side, the winner would end up with green, and the person who lost would be red.”

Student 4: “What about the wires?”

Humphrey: “The wires to some extent do have to be hidden, normally what we would have in competition when we have a setup like that is we would place the lighting system over that center target (refer to figure 2) and then we just throw the wires over the top. Most throwers won’t hit them.”

Student 4: “Do you guys have special thicker wires?”

Humphrey: “Uhm, I haven’t seen anything….If I was doing it, I would probably use a CAT 5 (type of wire, typically used for ethernet connection, and other uses). Because that would not only transmit the information for the LED, but you could also transmit the power for the lighting source. The trick here is that we probably would want to power it 12V, or using lithium ion batteries, with an adapter that we can run indoors when the power goes out.”

Student 1: “Sorry, you said green is you won, red is your lose, what would indicate to start throwing?”

Humphrey: “So the ready would be yellow, or amber, and green is actually the go throw, like the throw color. If you wanted, you could between the green, actually have the lights turn off and back on.”

Student 1: “Or like the winner stays green, and the loser turns red.”

Humphrey: “Ya, you could easily do it that way too. Just a couple less lines of code”

Student 3: “Does it have to be compatible with the other types of lights? Like for example, indoors you would want 5 light to be activated, and outdoors, you would want 3 to be activated?”

Humphrey: “So your talking in terms of brightness?”

Student 3: “No, for different lanes. For example, do you want it to be active for 3 of those?”

Humphrey: “Okay, for quick draw, it’s normally a head-to-head so 2 lanes. I have seen a lot of systems go up to 5 or 6 lanes for the gun ones, in our case, we don’t require that cause its a head-to-head battle.”

Student 1: "What would be a decent size light in your opinion, because I can see super small one that would be irritating for the eyes."

Humphrey: "Okay, let's put it simply, it has to be visible in sunlight, so whatever cluster it would have to be to be visible in sunlight."

Student 1: "And then what would be the costs. Like how much would you be willing to spend for this?"

Humphrey: "Okay, so commercially, these types of units range in price from around 800 to 3-4 thousand,"

Student 1: "For 2-lighting system?"

Humphrey: "For 2-lighting system depending on the builder. As I said, there isn't anything commercially ready for the knife throwing community. When I was doing my basic research, I figured, with all the little parts, parts are actually quite inexpensive. Parts would probably cost only 2-3 hundred dollars, not including some sort of enclosure to put it in, I'm not really counting that, I mean I could build it into a suitcase, or an old Xbox, doesn't really matter. Uhm, yeah the actual cost of the item isn't actually that expensive, but up until now, I guess people haven't really demanded it but I see kind of a lot of opportunity, when it comes to teaching, well even teaching kids, I think it would be an incredible activity and face off against each other, and kind of have fun that way. Uhm, and as I said, since they aren't too many of these systems out there, actually I did see a computer program recently that was written to do this, it was on instagram, but they had a laptop sitting between 2 axe throwing lanes, (laughs). Not the best idea. Ya so literally there is no fence here (referring to fence that separates 2 lanes), and you see the laptop right here (points to between the 2 lanes). So if they miss, its a laptop gone. I guess it's a business expense but still. I think having something that's decently modular would be great, so if your lighting systems and your timers are 2 separate pieces, and you have your main controller in a box that you can kind of tuck away somewhere, that'd be great."

Student 1: "So that you would say that the price does not matter, but you would prefer the price to be lower, right?"

Humphrey: "Well, …….yeah. Everyone would want the cheapest thing possible right? You want your best bank for buck. If I was to be making them and selling them, I would probably selling them to about 800 to 900 dollars, but it really depends on (what your bottom line is). So it really depends on what it cost to build in the first place."

Student 2: "so is there a specific deadline you are trying to make with these quick draw lights?"

Humphrey: “No, not of hand, I think Ryan and I have been playing with the idea for a while. I started research into it, just started researching to about a week ago. And started looking at coding of microcontrollers, stuff like that. C++ is just [mind blown]. So, like programing languages for me, I guess I could figure it out if I really spend the time, but at the end of the day, time is money too. So like, should I learn it myself, or should I find someone to code it for me, or what should I do, I guess I could just get my dad to code it if I really wanted to. He was a mainframe programer and I am sure he can figure it out but, you know, he like sitting at home playing with ipad all day long, so I don’t want to bother him either. But yeah no, I think there is kind of a lot of things would go into it, and so I don’t necessarily want to do it myself if I can.”

Student 1: “well you have us in your disposal”

……..

Humphrey: “It will push you towards electrical actually. You got your 5 volt system that controls everything, but most of your lighting would be like 12 volt, so you have to work in a system and bump it down for the microcontroller, and back up for your lighting and all that fun stuff. Not to mention figuring it out how to make it all work. Which I think conceptually it’s actually quite simple, it’s just that having the resources to put it together is another thing. And I don’t solter, I don’t have a breadboard, I don’t necessarily want to buy a breadboard to test it out. “
……..

Humphrey: “One of the interesting thing about this is the benefit to the community. There is actually a few communities that this would benefit. So you have your local throwing communities, so kids, their parents, etc, in the area. It provides them an opportunity for them to go head to head in a safe environment in a skill based activity that is physical, which is great. It actually promotes concentration as well as focus, which is great. Reflexes in the case of quick draw, definitely reflex. And not to mention comradery, and kind of a friendly competitive environment, it’s good to kind of compete. I mean everyone gets a participation ribbon, but sometimes winning is nice. And I mean as long as the coaching goes right, then losing isn’t necessarily a bad thing, right? I promotes trying harder or building the skills to do better. So that is a community that is, obviously, micro. On the macro side though, if we were be able to get a prototype or a working system in place, we could actually help throwers around the world. IKTHOF is a international community, that spans through Europe into North America. I am not even sure if the Asian areas do it, but I do know for a fact that a native american group here that actually practices here, goes to Korea every once in a while, they actually recently just went over to Korea to present their martial arts, and teach Koreans how to throw knife and spears and stuff like that. Which, you know, could also include us in a sense that we have a great system to work with.“

A typical example of what is used by the Stryke community in a quick draw competition. There would need a device to indicate to the player whether to throw or get ready to throw as well as who won.

3 Transcript of Interview #2 with employee, Michael

Feb 10, 2018
Stryke

looking at knives

Student 1: “So what are these holes for? (referring to circular dimples along the handle and blade of the knife”

Michael: “It’s for grip. And aesthetics too right.”

Student 1: “When you throw it do you hold from the handle or the blade?”

Michael: “It’s preference. There’s 4 different ranges you can throw from. There’s 2, 3, 4, 5 ranges and [the thrower] will change whether you’re holding it from the blade or handle. So there’s different ways to throw it as well. Some people hold it like a hammer. Others pinch throw it, there’s different variations So there’s a lot of variations with how you throw and choice of knives you use.”

...

Student 1: ”Can you write down the website [of where you get your knives from]. Is this the supplier you use?”

Michael: “No, so a lot of people buy from different websites like Amazon.”

...

Student 1: ”Is there like a quick draw competition but for axes?”

Michael: “Yeah it’s kinda the same thing. But that’s even more variable since people make their own axes. And they use different materials for handles. I believe they use different boards for axes.”

Michael takes out 3 different sized axes

Michael: “So there's varying handle length and it adds a little bit of weight to it. And people have difference preferences and all it does is change the rotation speed. The head is usually the same like stainless steel but the handles they use different types of wood so this is where it gets really annoying bc they use a lot of different types of wood, this changes the weight, they use a lot of handle length so that changes it up. Some people bolt [the head] down [to the handle] or they like tape it on.”

Student 1: ”There’s no point of weighing it cause the weights are so varying.”

...

Student 1: “So Humphrey was telling us in quick draw competitions they use a timer, when you throw the knife and it hits the board it senses it and stops the timer. But it’s designed for gun things - ”

Michael: “Yeah it’s not really meant for knife throwing.”

Student 1: ”What are the disadvantages of it not being specifically for knives and why does Humphrey want to make a new one”

Michael: “It’s more like it’s not as practical. The only reason I would see is that it’s doesn’t really really sense - it wouldn't be like super accurate. I guess when you’re using an accelerometer it’s more accurate. The second it impacts the board it stops the timer. I’m not sure how a gun accelerometer works so I can’t be completely sure.”

Student 1: ”Can you show us a quick draw demonstration?”

An instructor, Marcus, throws knives and axes while the 3 of us videotape it

...

Student 1: ”Do you know the grade of the wood you use for the targets?”

Michael: “Pine.”

Student 1: ”For every single target?”

Michael: “Yeah.”

Student 2: ”Do you know the grade of wood you use?”

Michael: “No.”

4 Transcript of Interview #3 with employee, Humphrey

Feb 12th, 2018 at Stryke

Humphrey: Using the center target on the 3-target setup, it's inside of the 5,4, and 3 rings, so that’s about an 8-inch octagon ring that you have to within, any of the outside areas, if your knife is in there, basically doesn’t count for anything

Student: And the timing mechanism, is it 3 to 8 seconds?

Humphrey: I believe it is 2 to 5 seconds, some random integer between 2 to 5 seconds.

Student: So actually, when we went online, we found that the gun one was 3 to 8 seconds, and I think it was pretty close to what you wanted. One thing is that we were wondering whether there are some issues with this, like the cowboy fast draw timer compared to what you guys wanted?

Humphrey: I think it's the kinetic force. For cowboy fast draw, those are design to be adhered to the back of a metal plate. So you are looking at the discharge of a firearm going into a metal plate, so the kinetic force…

Student: Would be much higher

Humphrey: A lot higher it might be of something getting a more sensitive sensor, but there is kind of a middle speed spot that I would assume that would work. On top of that, the cowboy quick draw system uses a much smaller timer system. I haven't seen a timer necessarily attached where it is visible to everyone around. I've seen the mastery units have timers on them, where they will tell you what times there are, which basically means that whoever is reffing would see the time, but everyone else wouldn't, so that's why I'm looking at the lighting system above the targets themselves, and have a giant LED readout of the times, you know, it would increase people's awareness so they know how fast things are throwing.

Student: Okay, and also, do you know where the quick draw is thrown from?

Humphrey: It's thrown from 3 meters roughly. Ya so 3 meters is the minimum distance, to wherever you are comfortable throwing a one-rotation throw. So in my case, I actually throw at 3 meters and probably 20-30 cm.

Student: Oh so just where you are comfortable?

Humphrey: Wherever you are comfortable, wherever you can make it work

Student: So 2 people don't have to start at the same position?

Humphrey: No

Student: Okay, interesting

Humphrey: That's why having the sensors on the board themselves would be ideal cause then you have the contact itself that will differentiate when the hits happen.

5 Transcript of Interview #4 with Assistant Manager, Humphrey

Feb 17th, 2018

Student 1: So, worst case scenario, would a competition go for 12h?

Humphrey: Typically in competition, you would normally have 30-40 throwers, its a best of 3 set. Normally, you would be running them 5 minutes a set, so you could run for about 2-3 hours tops, so you wouldn't have to worry about anything like battery life or anything like that.

Student 1: So we actually had another set of questions to ask, what are the tasks of the employees at Stryke?

Humphrey: So it depends on the role. All staff are expected to be able to coach, they must be decently proficient in axes, knives, archery, as well as guns, understanding maintenance for that equipment, as well as troubleshooting for the guns and archery specifically. They need to be able to see what is happening, and come up with a solution for that problem, so for example, if a thrower is under-rotating, they need to be able to have multiple ways of addressing that in order to get enough rotation, or give them enough space to make the rotation they require, so they have to have a few different tools to be able to address those types of complications. Beyond that, it's general cleaning up, and all the usual tasks as necessary kind of closet you would find in a typical contract of employment. Nothing obscene you know, we don't make anyone do anything against the law or anything like that, but its really that proficiency and those skills.

Student 1: And, who typically would make the boards?

Humphrey: It's basically an employee making the boards, he normally get 3-4 hour per week to replenish the boards for the weekends. He can pound out about 15 boards in a 4 hour shift.

Student 1: Wait so like for the employees, do we just have like regular employees? Or does everyone here have the same tasks?

Humphrey: Everyone essentially has the same tasks. People on con, so in my position are typically supervisors, so they do a little more logistical work, some administration, like I mean I handle the books and stuff, nothing obscene so I'm expected to be able to take a coaching shift, or a supervisor-y shift. And then there's Ryan, who's manager.

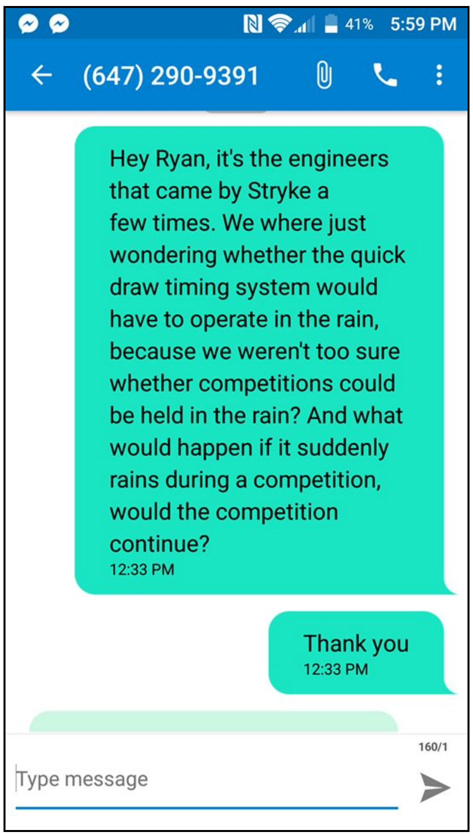
Student 1: Is he like head manager?

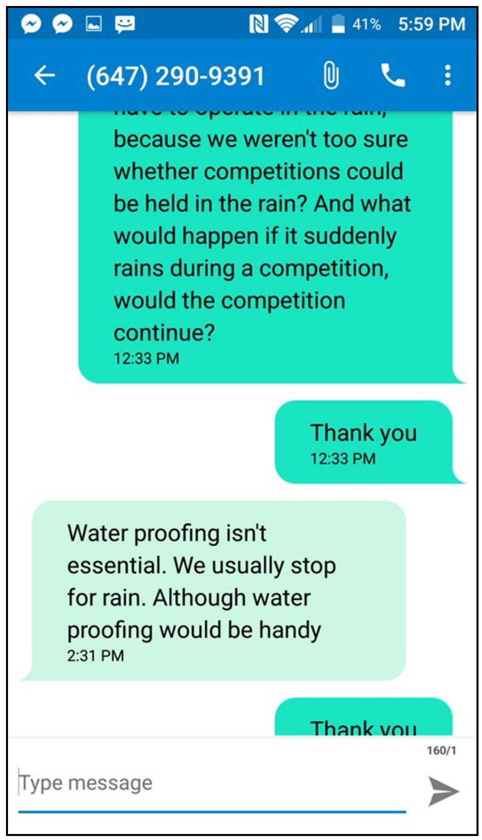
Humphrey: He is the manager for this location, there is only 1 manager.

Student 1: If we wanted this timer to be approved by the IKTHOF, who would we want to talk to?

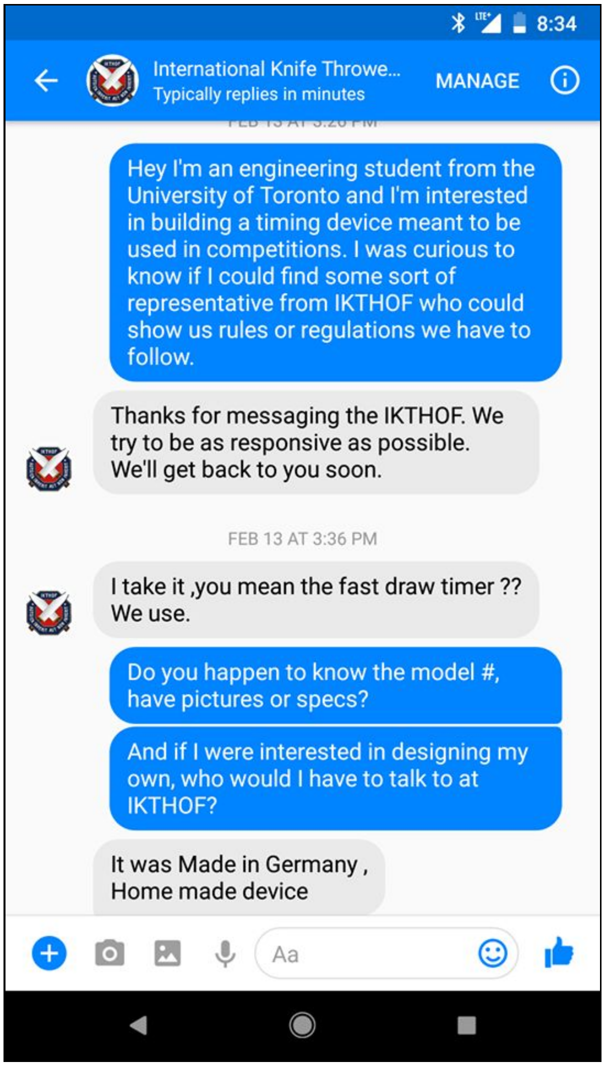
Humphrey: Uhm, the person who is in charge who is in charge of international is Ward Wright, you'll find him on the site, he's located in Pembroke, he's in the Ottawa area. He actually hosts his own throwing competitions in Ottawa, which I plan to go to in June. He does actually have one that was made for him, but as I explained to you guys before, it's tiny, and really hard to read, and the lighting system is not ideal

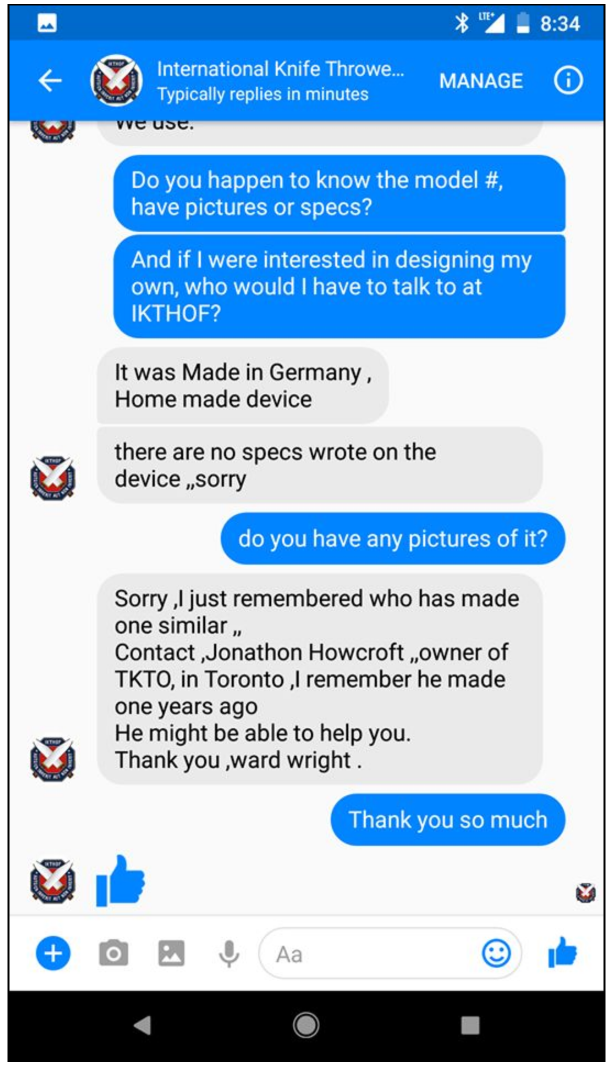
6. Text message with Stryke Manager Ryan





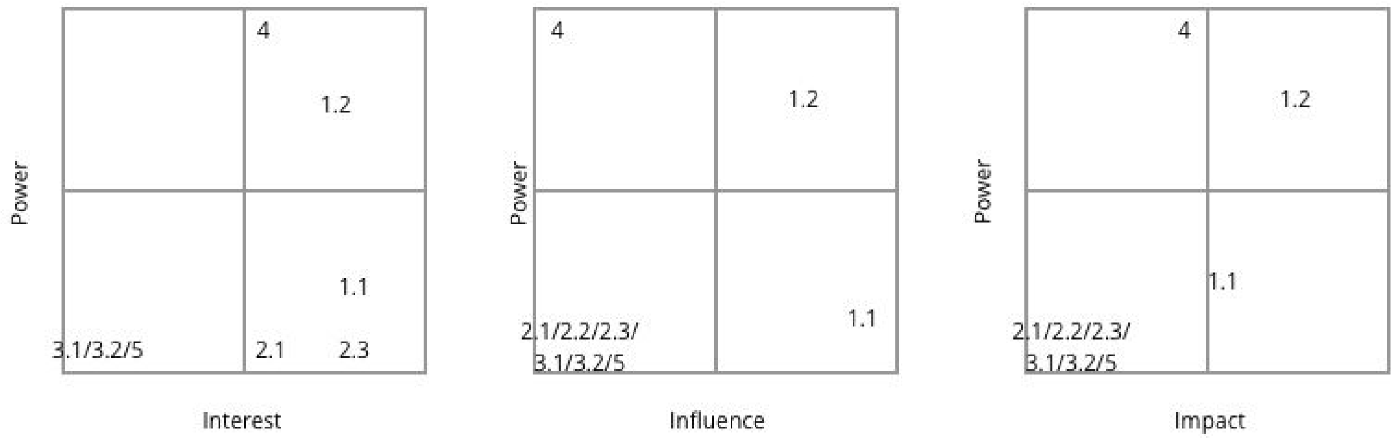
7 Facebook messages with Ward Wright, IKTHOF head of international affairs





Appendix C: Engineering Tools

1 Stakeholder Matrix



<u>Legend of Stakeholders</u>

1 Employees at Stryke

1.1 General Employees

Responsible for regular tasks such as cleaning, maintenance, equipment troubleshooting, coaching, and building targets at Stryke.

1.2 Manager

Responsible for administrative tasks at Stryke

2 Local throwing community

2.1 Advanced Thrower

Throwers with intention of improving their throwing skills

2.2 Leisure Thrower

Throwers with intention of enjoying the sport

2.3 Competitive and professional knife throwers

Throwers who participate in leagues and IKTHOF competitions

3 Competition Staff

Staff who operate knife throwing competitions

3.1 Setup crew

Responsible for setup of the device

3.2 Referee

Responsible for correctly identifying the winner

4 **IKTHOF** (International Knife Throwing Hall of Fame)

5 **Competition Audience**

As a result of the matrix the primary stakeholder were decided:

- IKTHOF
- The employees of Stryke

The rest are secondary stakeholders.